

Resting natural killer cells promote the progress of colon cancer liver metastasis by elevating tumor-derived sSCF

Reviewed Preprint

v2 • September 3, 2024

Revised by authors

Reviewed Preprint

v1 • June 11, 2024

Chenchen Mao, Yanyu Chen, Dong Xing, Teming Zhang, Yangxuan Lin, Cong Long, Jiaye Yu, Yunhui Luo, Tao Ming, Wangkai Xie, Zheng Han, Dianfeng Mei, Dan Xiang , Mingdong Lu , Xian Shen , Xiangyang Xue 

Department of General Surgery, The Second Affiliated Hospital and Yuying Children's Hospital of Wenzhou Medical University, Wenzhou, China • Department of Microbiology and Immunology, Institute of Molecular Virology and Immunology, School of Basic Medical Sciences, Wenzhou Medical University, Wenzhou, China • Department of Pediatric Thoracic Surgery, The Second Affiliated Hospital and Yuying Children's Hospital of Wenzhou Medical University, Wenzhou, China. • Department of Thoracic Surgery, The First Affiliated Hospital of Wenzhou Medical University, Wenzhou, China • Department of General Surgery, The First Affiliated Hospital of Wenzhou Medical University, Wenzhou, China

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

Purpose

The abundance and biological contribution of Natural killer (NK) cells in cancer are controversial. Here, we aim to uncover clinical relevance and cellular roles of NK cells in colon cancer liver metastasis (CCLM)

Methods

We integrated single-cell RNA sequencing, spatial transcriptomics, and bulk RNA-sequencing datasets to investigate NK cells' biological properties and functions in the microenvironment of primary and liver metastatic tumors. Results were validated through an in vitro co-culture experiment based on bioinformatics analysis.

Results

We used single-cell RNA sequencing and spatial transcriptomics to map the immune cellular landscape of colon cancer and well-matched liver metastatic cancer. We discovered that GZMK⁺ resting NK cells increased significantly in tumor tissues and were enriched in the tumor regions of both diseases. After combining bulk RNA and clinical data, we observed that these NK cell subsets contributed to a worse prognosis. Meanwhile, KIR2DL4⁺ activated NK cells exhibited the opposite position and relevance. Pseudotime cell trajectory analysis revealed the evolution of activated to resting NK cells. In vitro experiments further

confirmed that tumor-cell-co-cultured NK cells exhibited a decidual-like status, as evidenced by remarkable increasing CD9 expression. Functional experiments finally revealed that NK cells exhibited tumor-activating characteristics by promoting the dissociation of SCF (stem cell factor) on the tumor cells membrane depending on cell-to-cell interaction, as the supernatant of the co-culture system enhanced tumor progression.

Conclusion

Together, our findings revealed a population of protumorigenic NK cells that may be exploited for novel therapeutic strategies to improve therapeutic outcomes for patients with CCLM.

eLife assessment

This **useful** study draws on published single-cell and spatial transcriptomic data of colon cancer liver metastasis to clarify the pro- and anti-tumorigenic properties of NK cells. The authors discover increased GZMK⁺ resting NK cells in the tumor tissue and reduced abundance of KIR2DL4⁺ activated NK cells. However, the evidence is currently **incomplete**, as the models used to validate the hypothesis and claims are not adequate and lack the necessary controls.

<https://doi.org/10.7554/eLife.97201.2.sa3>

Introduction

Colon cancer is one of the most common malignancies, with more than 1 million new cases and 500 thousand deaths reported globally in 2020 (1). Approximately 19–26% of patients with colon cancer present with synchronous metastatic diseases at the first diagnosis (2), among which 14.5–17.5% develop liver metastasis (2, 3). Although progress in the treatment of metastatic disease, including improved diagnosis and treatment strategies for liver metastases (4–6), increased cancer-directed surgery (7), and the development of targeted therapies (8), has greatly improved the survival of these patients in recent decades, the 2-year relative survival rate for patients diagnosed with distant-stage disease was 36% (2), significantly lower than those without metastasis. Thus, colon cancer-derived liver metastasis (CCLM) is a clinical challenge that requires urgent development of novel treatment methods.

The tumor microenvironment (TME) is a dynamic environment that governs tumor behavior and is required for tumor cell survival, growth, proliferation, and metastasis (9). Unlike previous understanding, it has recently been proposed that certain immune cells present in the TME (such as myeloid-derived suppressor cells (10), macrophages (11), neutrophils (12), and CD8⁺ T cells (13)) favor tumor progression. Thus, the contradictory findings of the pro- and anti-tumor effects of tumor-infiltrating immune cells require a better understanding of the immune features and profile heterogeneity of colon cancer liver metastasis (CCLM) for the future development of immune-modulatory strategies to stratify and target immune cells.

Natural killer (NK) cells, an important component of tumor-infiltrating immune cells, are cytotoxic lymphocytes that play a key role in recognizing and eliminating malignant cells and are therefore considered the early responders against tumors (14). Among various immune cells, tumor-infiltrating NK cells are associated with prognosis in various cancers, such as colorectal cancer

(CRC) (15), lung cancer (16), and gastric cancer (17). Recently, based on their anti-tumor potential, NK cells were exploited for treating malignancies and have proven to be highly promising for the treatment of certain hematologic malignancies (18, 19) but have limited efficacy for treating solid tumors (20–22). Additionally, intratumor NK cells differ phenotypically or functionally from peripheral NK cells in several malignancies (23, 24); a specific subset of tumor-infiltrating NK cells (CD11b[−] and CD27[−]) as well as decidual-like NK (CD9⁺ and cd49a⁺) was associated with tumor progression in lung and hepatocellular cancers (25–27). Notably, the tumor-promoting transformation of NK cells may be educated by cancer cells (28, 29). However, their function remains elusive. In particular, the presence of specific NK cell subsets within colon cancer and their potential relationship to CCLM progression have not been fully characterized to date.

In this study, we combined previously published single-cell RNA sequencing, spatial transcriptomics (ST), and bulk RNA-sequencing data from public datasets, aiming to comprehensively chart the cellular landscape of TME in primary colon cancer and matched liver metastasis. Results of bioinformatics analysis were further validated through an in vitro co-culture experiment (Fig. 1A).

Materials and Methods

Data sources

In this study, we integrated four independent public datasets that contained single-cell RNA sequencing and ST data (10X Genomics) (derived from previously published research at the website: <http://www.cancerdiversity.asia/scCCLM/>) (30). And two bulk transcriptomics data of colon cancer (downloaded from TCGA cohort COAD at the website: <https://xena.ucsc.edu/> and NCBI's Gene Expression Omnibus with accession number GSE29623).

Single cell RNA Sequencing Analysis

The R (v4.0.5) package Seurat (v4.1.0) (31) was used to process the scRNA data. Since dataset quality control had been performed and Seurat objects had been created in previous studies, we did not further filter the scRNA-seq data or remove the batch effects. The SCTransform method was used to normalize the data. After selecting 2,000 highly variable genes using the Find Variable Features function in Seurat, principal component analysis was performed using these genes to reduce the data dimensions. Based on the ElbowPlot function in Seurat, we chose the top 30 principal components to run the FindNeighbors function. Next, the cells were clustered using the FindClusters function with a resolution of 0.1 for clustering immune cells. A uniform manifold approximation and projection algorithm was used for data visualization, as previously reported (32). Differentially expressed genes (DEGs) of each subset were identified using the FindAllMarkers function in Seurat. SingleR (v1.0.0) was used to name each cluster (33). Additionally, the subgroup of NK cells was defined by the preferential expression marker of the resting and activated NK cells described in the CIBERSORT.

Pseudotime Analysis

To analyze the differentiations of NK cells, monocle2 (<http://cole-trapnell-lab.github.io/monocle-release/>), which uses an algorithm to learn the changes in gene expression sequences that each cell must undergo as part of a dynamic biological process (34), was used for pseudotime trajectory analysis to identify the transitional relationships among different clusters. The cells were reduced dimensionally using the DDRTree method, sequenced in pseudotime, and finally visualized (35).

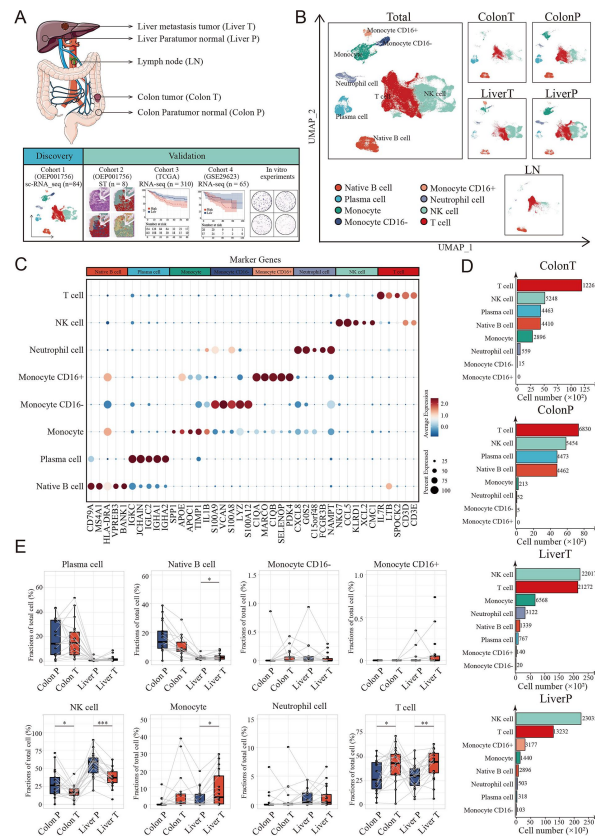


Figure 1.

Landscape of tumor immune microenvironment in CCLM revealed by single-cell transcriptomics

A. Schematic overview of the experimental design and analytical workflow. Colon T: colon tumor tissue; Colon P: colon paratumor normal tissue; Liver T: liver metastasis tumor tissue; Liver P: liver paratumor normal tissue; LN: lymph node metastasis tissue. B. The UMAP plot of all main immune cell types. C. Dot plots showing average expression of highly variable genes of each cell group. The dot size represents percent of cells expressing the genes in each cluster and the color of dot represents the expression intensity. D. The cell numbers of main immune cells across tissues. E. Proportions of all main immune cells. P values were determined by the paired nonparameter test.

Bulk RNA seq Data Analysis

The bulk transcriptome RNA-seq data and corresponding clinical data were obtained from The Cancer Genome Atlas of colon adenocarcinoma through the UCSC Xena browser (GDC hub) (<https://gdc.xenahubs.net>) (36). In total, 459 colon cancer samples with survival information were downloaded, and 310 samples were finally enrolled (11 formalin-fixed samples were excluded, and an additional 134 and 4 samples were excluded due to deletion of metastasis information and sequencing matrix, respectively). Transcriptomic data from 65 colon cancer samples in GSE29623 were obtained as the validation cohorts. CIBERSORT (37), xCELL (38), EPIC (39), and MCP-counter (40), which use gene expression to infer the proportions of tumor-infiltrating immune cells, were used to analyze the TME cell type. R packages survival (v3.2-10) and survminer (v0.4.9) were used for survival analysis. The Youden index was selected as the cutoff value to differentiate patients into distinct groups (high or low). The Kaplan–Meier survival curve was modeled using the survfit function. Subsequently, a Cox proportional hazards regression model was established to determine the independent risk factors. DEGs between the metastasis and non-metastasis groups were determined with the filtering condition of $\log_2 |\text{fold change}| > 1$ and $p\text{-value} < 0.05$, using the R package limma. GO and KEGG pathway functional enrichment analyses were performed using the clusterProfiler R package (v3.18.1) to assign various biological processes, molecular functions, cellular components, and pathways of identified marker genes in the cluster of interest (41), and $p < 0.05$ was regarded as statistically enriched. To explore the different KEGG pathways and hallmark gene sets between the metastasis and non-metastasis groups, gene set enrichment analysis (GSEA) was performed using data from The Molecular Signatures Database (c2.cp.kegg.v7.3.symbols) and the fgsea R package (v1.12.0) (42). Pathways with an adjusted p -value below 0.05 were deemed to be significantly enriched.

ST Data Analysis

Seurat was also used for ST data processing and visualization. We used the SCT method to normalize the ST data; the functions SelectIntegrationFeatures, PrepSCTIntegration, FindIntegrationAnchors, and IntegrateData were used to integrate the ST data. An unsupervised clustering method was subsequently used to cluster similar ST spots. Cell population annotations were based on hematoxylin and eosin (H&E) staining sections and the highly variable genes in each cluster. Scores of cell-specific signatures (top 20 DEGs) from scRNA-seq were calculated using the AddModuleScore function. SpatialDimPlot and SpatialFeaturePlot were combined to visualize the cell expression level in the ST data (32). Since the tumor and non-tumor region was identified by immunohistochemistry as well as cell clusters in ST data, the differences of cell subgroup presence in tumor and non-tumor region were determined by visual inspection.

Cell Lines and Co-cultures

Colon carcinoma cell lines HCT-116, DLD-1 and Human Foreskin Fibroblast (HFF) were purchased from the Cell Bank of the Chinese Academy of Sciences (Shanghai, China). HCT-116, DLD-1 were grown in a RPMI-1640 medium (GIBCO/BRL) supplemented with 10% FBS (GIBCO/BRL) while HFF were grown in the DMEM+10% FBS. The human NK cell line NK-92 was also purchased from the Cell Bank of the Chinese Academy of Sciences (Shanghai, China) and maintained in RPMI-1640 supplemented with 12.5% horse serum (GIBCO), 12.5% FBS, 100 U/mL rhIL-2, 0.1 mmol/L β -mercaptoethanol, and 0.02 mmol/L folic acid. All the cells were cultured in a humidified chamber at 37 °C and 5% CO₂.

Colon cancer cells were plated at a density of 1×10^5 cells/well in 6-well plates for 24 h. Supernatants were collected and cells and debris therein were removed by centrifugation. NK cells were co-cultured with cancer cells in a ratio of 1:1 in a fresh mixed medium for a further 24h (CN group). Additionally, NK cells were also cultured in a mixed medium of supernatant and NK medium in the ratio 1:1 (SN group) as well as fresh RPMI-1640 10% FBS and NK medium in the

same ratio (MN group) for 24h (Fig.S1A). When indicated, co-cultures were performed in Transwell devices (JET biofil), maintaining the same ratios and culture times. Finally, as shown in Fig.S1B, the co-cultured supernatant of CN, SN and MN group were collected, which determined as CNS, SNS and MNS, to incubate fresh HCT-116 to explore the NK cell-mediated inductive effect on colon cells.

Flow Cytometry Analysis

NK cells in each co-cultured group were collected and washed twice in $1 \times$ PBA. Then, they were incubated with APC-labeled CD56 (Biolegend), PE-labeled KIR2DL4 (Affinity Biosciences), FITC-labeled GZMK (Affinity Biosciences), FITC-labeled CD9 (Biolegend), PE-labeled PD-1 (Biolegend) and PerCP-labeled CD49a antibody for 30 min, rewashed, and resuspended in PBS. Fluorescence Minus One (FMO) control was set up to define the position of the positive gate as shown in Table. S1 and Table. S2. BD flow cytometry was used for detection.

Cell Counting Kit-8 (CCK-8) Cell Viability and Cell Colony Formation Assay

Colon cancer cells (5,000 cells/well) were cultured in 96-well plates with co-cultured supernatants from different co-culture groups alone or with both co-cultured supernatants and imatinib mesylate (2 μ M) as previously reported ([43](#)) for 24 h. At pre-determined time points, 10 μ L of CCK-8 reagent (Dojindo, Japan) was added and incubated for 2 h at 37 °C, and then the absorbance was measured using a microplate reader (Thermo Scientific) at 450 nm. All experiments were carried out in triplicate.

For the cell colony formation assay, colon cancer cells (500 cells/well) were seeded in 12LJwell plates with co-cultured supernatants alone or with both co-cultured supernatants and imatinib mesylate at 37 °C for 1 week. Then, cell colonies were fixed with 4% paraformaldehyde for 10 min and stained with 0.5% crystal violet for 5 min. Cell colonies containing >20 cells were counted. All experiments were carried out in triplicate.

Transwell Migration/Invasion Assays

The polycarbonate membrane in the transwell chambers was coated with Matrigel (Corning, NY, USA). Next, we transferred 1×10^5 cells from the serum-free medium with or without imatinib mesylate into the top chamber, added co-cultured supernatants alone or with both co-cultured supernatants and imatinib mesylate in the bottom chamber, and incubated at 37 °C for 24 h. Then, we removed the non-invading cells on the top side of the membrane by scrubbing, fixed the migrating or invading cells at the bottom side of the membrane with 4% paraformaldehyde, and stained with 0.5% crystal violet. The number of cells was counted under a microscope (Leica, London, UK) from four randomly chosen fields per well to determine the number of cells in each group.

Total RNA Extraction and Quantitative Real-Time PCR (qPCR)

Total RNA was extracted from cells using TRIZOL reagent (Invitrogen, USA). RNA was reverse transcribed into cDNA using a PrimeScript RT Reagent Kit (Takara, Japan). qPCR was performed using QuantStudio™ Test Development Software (Thermo Scientific, Waltham, MA, USA) with SYBR Green qPCR Master Mix (EZBioscience, Roseville, MN, USA). The sequence of KITLG and housekeeping gene GAPDH primers is listed in Table S3. KITLG mRNA data were normalized to that of GAPDH.

Luminex Liquid Suspension Chip Detection and ELISA

Luminex liquid suspension chip detection was performed by Wayen Biotechnologies (Shanghai, China). The Bio-Plex Pro Human Chemokine Panel 48-plex kit was used in accordance with the manufacturer's instructions. Briefly, supernatants from different co-cultured groups were

incubated in 96-well plates embedded with microbeads for 1 h and then incubated with a detection antibody for 30 min. Subsequently, streptavidin-PE was added in each well for 10 min, and values were read using the BioPlex MAGPIX System (Bio-Rad). The Human SCF ELISA kit (Solarbio, China) was used according to the manufacturer's instructions.

Statistical Analysis and Visualization

All statistical analyses were performed using SPSS (v23.0; IBM SPSS, Chicago, IL, USA) and R (v4.0.5), and data visualization was performed on R packages Seurat (v4.1.0), ggplot2 (v3.3.5), ggsignif (v0.6.1), and pheatmap (v1.0.12).

Results

Integrated scRNA-seq and ST Precisely Quantified Immune Cell Diversity in CCLM

We used a previously published single-cell dataset containing 89 samples from paired samples of colon cancer, adjacent colon, liver metastasis, adjacent liver, lymph nodes along colons, and peripheral blood mononuclear cells from 20 patients to define the heterogeneous immune microenvironment landscapes of CCLM. Subsequently, 178,630 CD45⁺ cells were integrated.

To further define the main cell type, clustering analysis and SingleR were performed, and monocytes, neutrophil cells, native B cells, plasma cells, T cells, and NK cells were identified from the CCLM samples (**Fig. 1B-C**; **Fig. S2A and B**; Table.S4). Notably, T and NK cells were the major components of all immune cells and distinct across different tissue types, with NK cells significantly decreasing in tumor tissues (both primary colon cancer and liver metastasis cancer) compared to corresponding adjacent normal tissues, whereas T cells were the opposite (**Fig. 1D** and **E**; **Fig. S2C**).

To comprehensively analyze the spatial distribution profile of colon primary tumors and CCLM tumors, we collected ST data, which came from eight samples including paired primary colon cancer and liver metastasis. Unsupervised clustering analysis was first performed to cluster similar ST spots, and the annotation of the clusters was further determined according to cell marker genes (**Fig. S3A-C**, Table.S5). Further, after combining the gene expression features of each sample (**Fig. S4**) and H&E staining, six morphological regions, including tumor, fibroblast, smooth muscle, B cells, hepatocytes, normal epithelium, and tumor and paratumor areas, were identified. (**Fig. 2A-C**).

To integrate the scRNA-seq and ST data, we used the AddModuleScore function in Seurat to quantify the main immune subpopulations. Consistent with scRNA-seq, T cells were remarkably enriched in the tumor region of both primary colon cancer and liver metastasis cancer, whereas NK cells were enriched in the non-tumor area (**Fig. 2D**).

Biological Relationship Between NK Cells and Metastasis of Colon Cancer Revealed by Bulk RNA Transcriptomics

TME cell composition differences between metastasis and non-metastasis colon cancer were evaluated using multiple tools that could robustly quantify the abundance of cell populations based on transcriptomic datasets, including xCell, EPIC, quanTIseq, and MCPcounter and bulk-seq datasets (TCGA COAD cohort). The percentages of NK cells markedly decreased in metastasis colon cancer using MCPcounter, EPIC, and xCell (**Fig. 3A-D**).

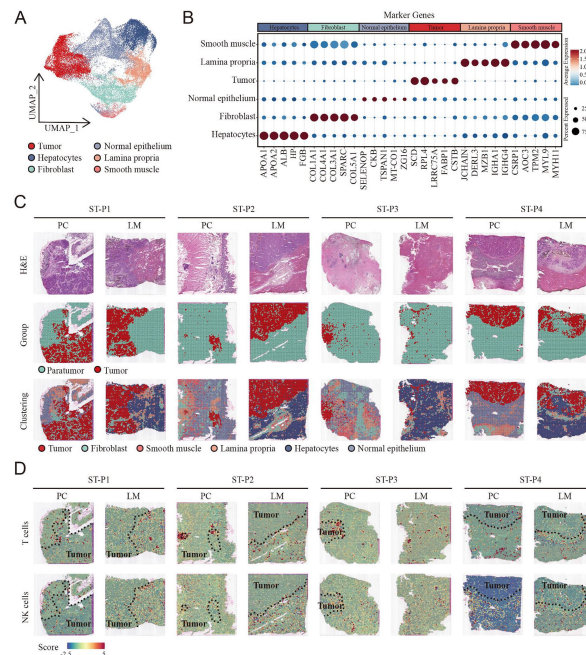


Figure 2.

Cellular identification in spatial transcriptomic samples

A. UMAP visualization of cell clusters in spatial transcriptomic samples. B. Dot plots showing average expression of markers in indicated cell clusters. C. Overview of the spatial transcriptomic sections. H&E staining of spatial transcriptomic sections (upper). Tumor tissue and paratumor tissue identification of each section (middle). Spatial cluster distribution of each section (lower). PC: primary cancer; LM: liver metastasis. ST-P1: spatial transcriptomic dataset patient 1; ST-P2: spatial transcriptomic dataset patient 2; ST-P3: spatial transcriptomic dataset patient 3; ST-P4: spatial transcriptomic dataset patient 4. D. The signature scores of T cells (upper) and NK cells (lower) in colon cancer and liver metastasis in the spatial transcriptomic sections.

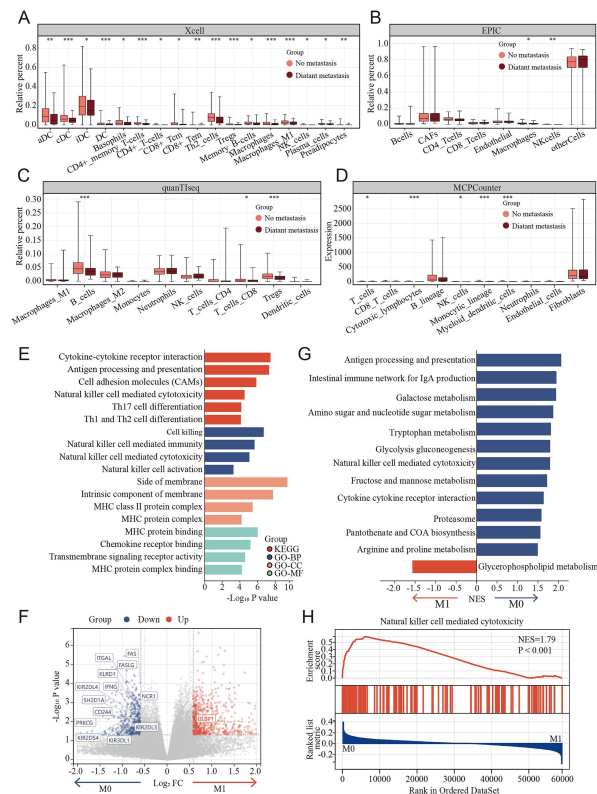


Figure 3.

Clinical and biological relationship between NK cells and metastasis of colon cancer revealed by Bulk RNA transcriptomics

A-D. The relationship of immune cell percentage determined by xCell, EPIC, quantiseq and MCPCounter between metastasis and non-metastasis tumor in TCGA COAD cohort. E. Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enriched pathway bar chart of Differentially expressed genes (DEGs) in metastasis versus non-metastasis colon cancer. F. Volcano plot showing nearly all the genes enriched in pathway of NK cell-mediated cytotoxicity were downregulated in the metastasis colon cancer. G. Gene set enrichment analysis (GSEA) of KEGG gene set. H. Natural killer cell mediated cytotoxicity was enriched in the non-metastasis colon cancer.

To further understand the potential triggers that induce colon cancer metastasis, the DEGs were calculated between the metastasis and non-metastasis colon cancer groups in the TCGA COAD cohort. A total of 1,378 DEGs were determined using the limma package with cutoffs $|\log \text{ fold change}| > 1.5$ and $p < 0.05$, including 817 upregulated and 561 downregulated genes between metastasis and non-metastasis colon cancer. Subsequently, Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analyses were performed to investigate the correlative functions and pathways. As shown in **Fig. 3E**, cell killing, NK cell-mediated immunity, NK cell-mediated cytotoxicity, NK cell activation, and MHC-related GO terms and the KEGG pathways of NK cell-mediated cytotoxicity were significantly enriched. Interestingly, nearly all the genes enriched in pathway of NK cell-mediated cytotoxicity were markedly downregulated in the metastasis group compared to the non-metastasis group (**Fig. 3F**). Similarly, Gene Set Enrichment Analysis (GSEA) revealed that NK cell-mediated cytotoxicity was enriched in the non-metastasis group (**Fig. 3G** and **H**).

The Landscape of NK Cells in CCLM Progression

Considering the differences in infiltrated NK cells between tumor-adjacent tissues and tumor tissues as well as colon cancer with and without metastasis, we further focused on NK cells to explore the detailed difference of NK subgroups in the CCLM microenvironments. Unsupervised clustering analysis of NK cells was thus performed and Eight NK cell subtypes were thus identified (**Fig. 4A**). Combining the highly variable features of each NK cluster (**Fig. 4B**) and CIBERSORT-reported canonical NK cell markers (**Fig. 4C** and **D**), NK cells were classified into three cell types, including activated NK cells identified by the expression of KIR2DL4, GPR183, GRP171, CD69, and IFNG, resting NK cells marked by GZMK, TTC38, CD160, and PLEKHF1, and the other NK cells of which no characteristic gene was identified. Although all three NK cell subtypes were presented in primary colon tumors, adjacent normal colon tissues, liver metastasis tumors, and lymph nodes (**Fig. 4E**), the infiltration grade for each of these NK cell subsets was significantly different. At the individual sample level, considerable variability was observed in the NK cell subset composition. Where serial samples were available from individual patients, the expression of resting NK cells increased, whereas that of activated NK cells decreased during disease progression (**Fig. 4F** and **G**). Additionally, using the paired primary colon tumor, adjacent normal colon tissues, and liver metastasis tumor samples, we observed that the gradient of the resting NK cells increased, whereas the gradient of the activated NK cells decreased (**Fig. 4H**).

To evaluate the clinical relationship between NK cell differences and CCLM, CIBERSORT was performed using the TCGA COAD cohort. The percentages of resting NK cells significantly increased (**Fig. 5A**), and the activated NK cells decreased in the metastasis group (**Fig. 5B**), which is consistent with the scRNA-seq results. Also, we observed that the resting NK cells were significantly increased, whereas the activated NK cells decreased in higher T, N, and TNM stages (**Fig. 5S**). In terms of survival, neutrophils and resting NK cells were associated with a worse prognosis, whereas activated NK cells and M1 macrophages were associated with better outcomes (**Fig. 5SB**). Further, survival analysis showed that colon cancer patients with lower and higher degrees of infiltration of activated NK cells and resting NK cells, respectively, had a significantly worse prognosis in the TCGA COAD and GSE29623 cohorts (**Fig. 5C** and **D**). Interestingly, resting NK cells were remarkably enriched in the tumor region, whereas activated NK cells were enriched in the non-tumor area of both primary colon cancer and liver metastasis (**Fig. 5E**).

Characterization and Developmental Course of Differential Subsets of NK Cells in CCLM

To further study the heterogeneity of NK cells subgroup, the proportions of all eight clusters of NK cells were calculated (**Fig. 56**). KIR2DL4+ GPR171+ activated NK cells were lessened in proportion with the progression of the disease from normal colon to colon cancer and liver metastasis cancer, whereas GZMK+ resting NK cells increased (**Fig. 6A** and **B**), suggesting that

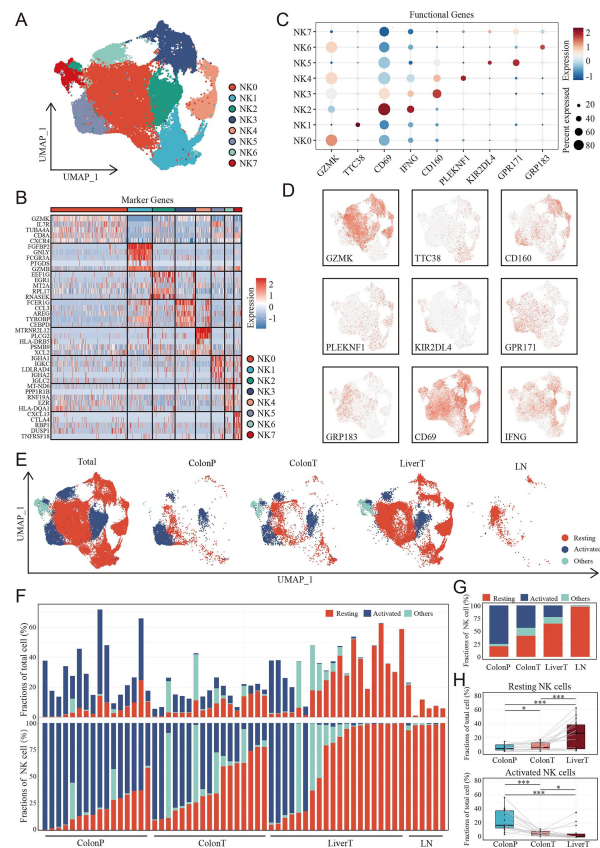


Figure 4

The landscape of NK cells in the disease progression of CCLM

A. The UMAP plot of NK cells from CCLM. B. Unsupervised clustering identifies 8 subsets of NK cells. C. Expression of key markers that distinguish resting and activated subsets of NK cells. D. Expression of key resting and activated NK cell markers across all samples. E. The UMAP plot of distribution of resting and activated NK cells from ColonP, ColonT, LiverT and LN. F. Cellular landscape of each sample from the ColonP, ColonT, LiverT and LN. The proportion of NK cells subsets in total immune cells (upper). The proportion of NK cells subsets in total NK cells (lower). G. Number of cells identified from each group (ColonP, ColonT, LiverT and LN) by cell type proportion. H. Proportions of resting (upper) and activated (lower) subsets of NK cells. P values were determined by the paired nonparameter test.

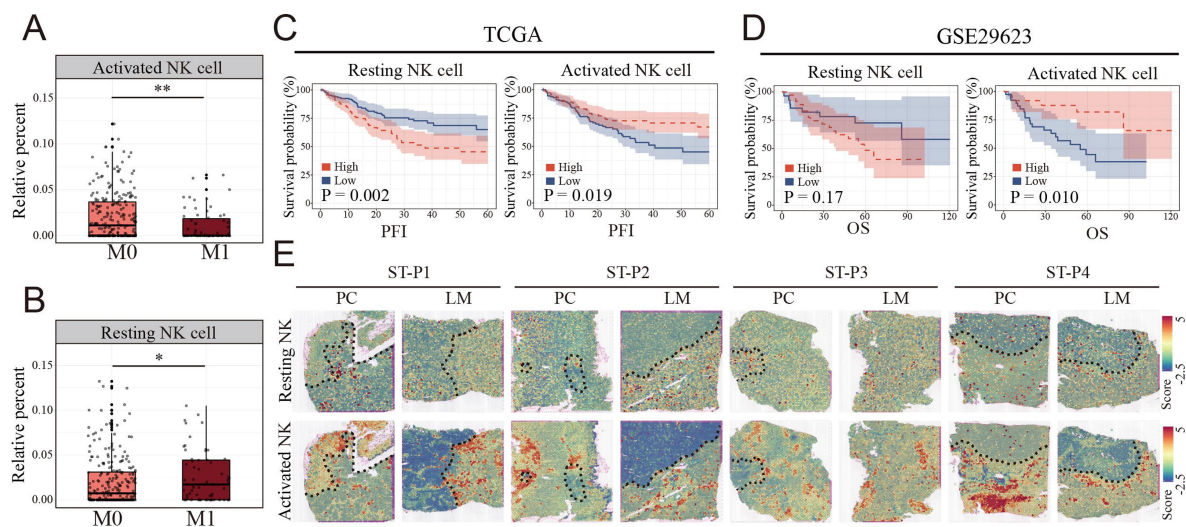


Figure.5

Clinical relationship between NK cells subsets and metastasis of colon cancer revealed by Bulk RNA and spatial transcriptomics

A-B. The relationship of activated and resting NK cell percentage determined by CIBERSORT and tumor metastasis in TCGA COAD cohort. M0: non-metastasis colon cancer; M1: metastasis colon cancer. C-D. K-M survival plots show that high resting NK cell and low activated NK cell predicted poor prognosis in TCGA COAD and GSE29623 cohort. PFI: Progression Free Interval; OS: Overall Survival. E. The signature scores of resting (upper) and activated NK cells (lower) in colon cancer and liver metastasis in the spatial transcriptomic sections.

these two NK subsets were associated with metastasis. ST analysis also confirmed that there was more infiltration of the KIR2DL4+ GPR171+ activated NK cell subset in the non-tumor area and GZMK+ resting NK cells in the tumor region, both primary colon cancer and liver metastasis (**Fig. 6C** [↗](#)).

Pseudotime cell trajectory analysis of the two NK cell clusters was constructed to investigate the evolutionary dynamics of metastasis-associated NK cells. NK cells from the normal colon group were located at the top right corner of the trajectory curve, suggesting the clear starting point of this evolving trajectory curve. After confirmation of this starting point, developmental routes were determined to begin with the KIR2DL4+ GPR171+ activated NK cells and then develop into GZMK+ resting NK cells (**Fig. 6D** [↗](#)).

To explore the transitional relationships of marker genes in NK cells in distinct clusters, we examined the expression level of these markers in different clusters through pseudotime analysis. Marker genes of activated NK cells (KIR2DL4 and GPR171) exhibited a gradient descent expression pattern, whereas the expression of resting NK cell marker genes (GZMK) exhibited a gradient rise pattern (**Fig. 6E** [↗](#)). BEAM analysis was performed to identify the cell cluster marker genes that change as NK cell subtypes move from the activated stage to the resting stage. The branched heatmap showed the expression dynamics of the top 200 significant genes in different cell fate branches. Corresponding GO enrichment analyses of these significant genes further demonstrated that a mass of leukocyte differentiation and immune system process-related GO terms were significantly enriched (**Fig. 6F** [↗](#)).

Interestingly, patients with colon cancer with higher infiltration of GZMK+ resting NK cells (**Fig. 6G** [↗](#)) and lower infiltration of KIR2DL4+ GPR171+ activated NK cells (**Fig. 6H** [↗](#)) exhibited shorter survival in the TCGA COAD cohort.

Tumor Cells-Educated NK Cells Shift Toward Tumor-Promoting Status Depends on Cell-to-Cell Interaction

To verify the NK cell-mediated inductive effect on colon cells, we set up mixed cell co-culture experiments using colon cell line HCT-116 and NK cell line NK-92 in the ratio 1:1 (CN group). As controls, NK cells were cultured alone, either in the tumor supernatant (SN group) or fresh medium (MN group). The co-cultured supernatant of each group were collected to incubate fresh HCT-116 after 24 h of co-culturing. Interestingly, CCK-8 and colony formation assays showed that colon cells in CNS group, which colon cells were cultured in the supernatant of co-cultured system that NK and cancer cell were in contact, underwent a significant increase in proliferation (**Fig. 7A-C** [↗](#)) compared to the CS group, where cancer cells were cultured directly in the supernatant of cancer cell after 24h culturing. Similarly, the migration and invasion of HCT-116 were also significantly increased in CNS group (**Fig. 7D-F** [↗](#)).

Furthermore, colony formation assays (**Fig. 7G-H** [↗](#)), CCK-8 (**Fig. 7I** [↗](#)), and transwell assays (**Fig. 7J-L** [↗](#)) showed colon cells in CNS group underwent a significant increase in proliferation, invasion and migration, compared to those cultured in the co-cultured supernatant (SNS group) and fresh medium (MNS group).

Additionally, another colon cancer cell line, DLD-1 was chosen to evaluate the effect of the supernatant in the different co-culture groups. DLD-1 cells in the CNS group did not undergo a prominent increase in proliferation compared to CS group (**Fig. S7A-C** [↗](#)). The same results were obtained for their migration and invasion (**Fig. S7D-F** [↗](#)). Similarly, no such cell functional changes were induced in the CNS group compared to SNS and MNS groups (**Fig. S7G-L** [↗](#)).

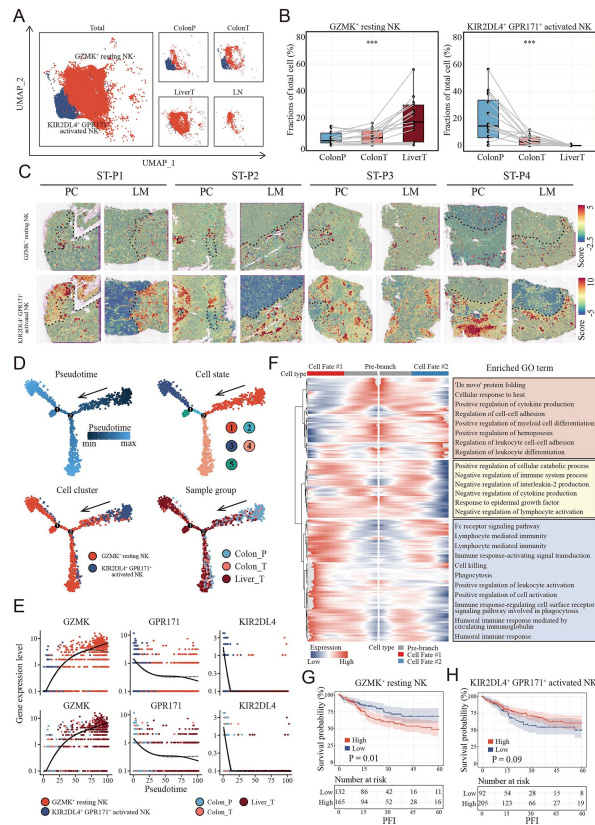


Figure 6.

Characterization and developmental course of differential subsets of NK cells in CCLM

A. The UMAP plot of KIR2DL4⁺ GPR171⁺ Activated NK cells, GZMK⁺ resting NK cells from CCLM. B. Proportions of KIR2DL4⁺ GPR171⁺ Activated NK cells, GZMK⁺ resting NK cells in total immune cells. P values were determined by the paired nonparameter test. C. The signature scores of GZMK⁺ resting NK cells (upper) and KIR2DL4⁺ GPR171⁺ Activated NK cells (lower) in colon cancer and liver metastasis in the spatial transcriptomic sections. D. Monocle analysis showing the developmental trajectory of NK cells. Color as in Pseudotime, cell state, subsets of NK cells and sample group. E. PseudoLJtemporal change curve of marker genes in each subsets of NK cells. F. The heatmap shows the expression patterns of the top 50 significant genes in branched expression analysis modeling, associated GO terms (using DAVID v6.7) are given on the right of the corresponding gene clusters. G-H. The Kaplan-Meier curve shows COAD patients survival with different GZMK⁺ resting NK cells and KIR2DL4⁺ GPR171⁺ Activated NK cells infiltration.

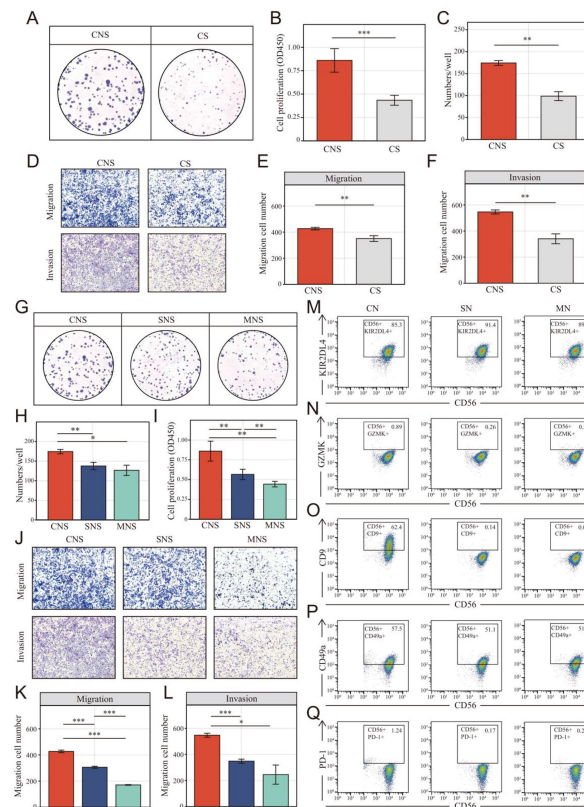


Figure.7

Colon cancer cells (HCT-116) educated NK cells shift toward tumor promoting status depends on cell-to-cell interaction

A-B. Clone formation assay showed the NK cell-mediated inductive effect on cell proliferation of colon cancer cell (HCT-116). Colon cancer cells were cultured in the supernatant from different co-culture system in transwell devices. CNS: colon cancer cells were cultured in the supernatant from co-culture system that NK cells and cancer cells were cultured in direct contact (CNS); CS: cancer cells cultured directly in supernatant of cancer cells. C. Cell Counting Kit-8 (CCK-8) assay showed the NK cell-mediated inductive effect on cell proliferation of colon cancer cell. D-F: The NK cell-mediated inductive effect on migration and invasion of colon cancer cell. G-H: Clone formation assay CCK-8 assay showed the NK cell-mediated inductive effect on cell proliferation of colon cancer cell. SNS: colon cancer cells were cultured in the supernatant from co-culture system that NK cells were cultured in supernatant of cancer cells; MNS: colon cancer cells were cultured in the supernatant from co-culture system that NK cells were cultured in fresh medium. M-Q. Phenotype switch (KIR2DL4, GZMK, CD9, CD49a, PD-1) of NK cells was induced by cell-to-cell interactions with cancer cells. CN: NK cells were co-cultured with colon cancer cells; SN: NK cells were cultured in supernatant of cancer cells; MN: NK cells were cultured in fresh medium.

Given the characteristic spatial distribution of NK cell subpopulations, we hypothesized that the tumor region-enriched NK cells might be educated by tumor cells and thus functionally distinct. To verify the possible effect of tumor cells on the phenotypic switch of NK cells, NK cells were analyzed using FACS to determine the expression of the NK cells markers above identified (GZMK and KIR2DL4) as well as tumor infiltrated NK cells markers: CD9, CD49a and PD-1. FMO control was set up to define the position of the positive gate as shown in Fig.S8. Upon exposure to tumor cells, NK cells significantly decreased and slightly increased the expression of activated marker KIR2DL4 (**Fig. 7M**) and the resting marker GZMK (**Fig. 7N**), respectively. On the other way, for GZMK, which known as the secretory protein, flow cytometric analyses were also performed with cell fixation and permeabilization. However, nearly all the NK cells in different groups were GZMK positive and no significant differences were found among each group (Fig.S9). Additionally, NK cells in CN group underwent an increasing of tumor infiltrated NK cells markers, CD9 (**Fig. 7O**), CD49a (**Fig. 7P**) and PD-1(**Fig. 7Q**), especially CD9. However, no such effect was observed after co-culturing with tumor supernatant and fresh medium.

On the other way, NK cells were also co-cultured with HFF, a normal human cell line. The tissue infiltrated NK phenotype identified above (CD9, CD49a, PD-1) were determined. When co-cultured with HFF in direct contact (CN group), NK cells were also tending towards tissue infiltration state. However, the domestication effect was significantly reduced compared to co-culturing with tumor cells (Fig.S10A and B). Additionally, fresh HCT underwent a limited increase (no statistical significance was found) in migration when cultured in CNS group, but not in the SNS group MNS group (Fig.S10C-E).

Resting NK-induced Colon Cancer Malignant Phenotype Promotion Depends on SCF Release

To study the potential mechanism by which the supernatant from the NK and colon cells in direct contact with the co-culture system promoted colon cell migration, Luminex liquid suspension chip detection was used to compare the differential expression of 48 common chemotactic and inflammatory cytokines in the CNS, SNS, and MNS groups. We observed that SCF was upregulated in all three repetitions in the CNS group (**Fig. 8A-B**). Further, ELISA confirmed the upregulated level of SCF in the CNS group (**Fig. 8C**).

We further assessed the inhibitory effects of imatinib mesylate on the enhanced proliferation and invasion. CCK-8 (**Fig. 8D-E**) and colony formation assays (**Fig. 8F-I**) showed that imatinib mesylate (2 μ M) significantly inhibited supernatant-enhanced proliferation of HCT-116. Similarly, the supernatant-enhance migration (**Fig. 8J-K**) and invasion (**Fig. 8L-M**) were also inhibited by imatinib.

Discussion

CCLM is a multistep process and is mostly fatal (44). During disease progression, functional interactions between tumor cells and the surrounding TME are critical for influencing tumor growth, promoting angiogenesis, and finally resulting in metastasis (45). Since the genomic divergence between primary and metastatic CRC cells is relatively low (46, 47) and accumulating evidence has highlighted the key role of tumor-infiltrating immune cells in dictating the fate of cancer cells (48, 49), we presented a comprehensive cellular and spatial immune landscape of the primary and liver metastatic tumors of CRC by using previously published scRNA-seq and ST data. Consistent with previous studies (30, 50), we observed that the immune microenvironment of primary and liver metastasis lesions of colon cancer had undergone extensive remodeling with a significantly decreased proportion of NK cells and a strong

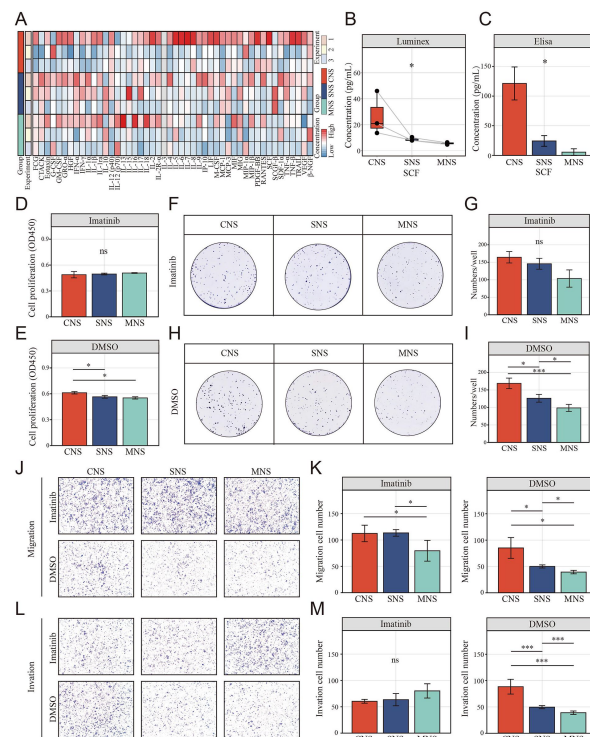


Figure.8

The resting NK cell promote tumor malignant phenotype via elevating tumor-derived sSCF

A. luminex liquid suspension chip detection of 48 common chemotactic and inflammatory cytokines in CNS, SNS and MNS group. B-C. Concentration of SCF determined by luminex liquid suspension chip and Elisa in CNS, SNS and MNS group. D-E. CCK-8 assay showed the proliferation of HCT-116 cells was inhibited by imatinib mesylate, evaluated by a CCK-8 assay. Cells were incubated in the supernatant from different co-culture system with DMSO or 2 μ M imatinib mesylate. F-I. Clone formation assay showed the proliferation of HCT-116 cells was inhibited by imatinib mesylate. J-M. The NK cell-mediated inductive effect on migration and invasion of HCT-116 was inhibited by imatinib mesylate.

enrichment of T cells. However, the contrast between the decreased proportion of NK cells in tumor tissue and the enrichment of NK cells in tumor regions may indicate the specific effects of NK cells in CCLM.

Traditionally, NK cells within the TME were known as potential suppressors for tumor growth by directly killing cells and secreting proinflammatory cytokines (51 [↗](#), 52 [↗](#)). However, the function of NK cells highly depends on their maturation status and localization (53 [↗](#)). The peripheral blood CD56^{dim}CD16^{high} NK cell population predominantly mediates the killing of target cells by secreting perforin and granzymes (54 [↗](#)), whereas the CD56^{bright}CD16^{low} NK cells that reside in secondary lymphoid tissues are immature and have reduced cytotoxic capability (55 [↗](#)). Recently, tumor-associated NK cells, which are enriched in tumors with impaired anti-tumor functions, were discovered to be associated with an unfavorable prognosis and resistance to immunotherapy in multiple solid tumors (56 [↗](#)), such as lung cancer, pancreatic cancer, and esophageal cancer. The potential impact of NK cells on CCLM progression is still poorly investigated.

In this study, the proportion of NK cells was higher than in past cognition, similar to a previous study (57 [↗](#)) in that NK cells occupied the largest immune cell compartment of nearly 50% in cholangiocarcinoma. In contrast, NK cells may not be well distinguished during automatic annotation using SingleR. Consistent with the study that revealed that treatment responders are associated with TME remodeling, including NK cell recruitment (58 [↗](#)), the proportions of NK cells in PR patients were significantly higher among the LiverP and LiverT groups (**Fig. S2D and E** [↗](#)), which also increased the overall proportion of NK cells to some extent. Additionally, we observed that NK cells are decreased in primary colon cancer and liver metastatic using scRNA-seq, consistent with previous findings supporting that NK cells play a crucial role in anticancer immunity (59 [↗](#), 60 [↗](#)). However, we further subdivided NK cells considering the conflicting result, in which NK cells were significantly enriched in the tumor area when using ST. Since this is the first study to comprehensively explore the heterogeneity of NK cells, no previous NK cell subpopulation markers were available; resting and activated NK cell markers in CIBERSORT were thus used to annotate NK cells. Consistent with a recent study that reported that mature and immature NK cells serve different functions (61 [↗](#)), our results also provide evidence that resting NK cells in the colon cancer TME possess a potential tumor-promoting activity, whereas activated NK cells play an anti-tumor role as conventional cognition. Bulk RNA-sequencing and the corresponding clinical data further indicated that higher infiltration of resting NK cells and lower infiltration of activated NK cells were correlated with a worse prognosis in patients with CRC. However, since the resting and activated NK cells we identified were only the aggregation of multiple NK cell subpopulations, further studies are required to investigate the key NK cell subpopulations that play a crucial role in CCLM.

Further analysis of NK cell subtypes revealed a dramatic change in KIR2DL4⁺ GPR171⁺ activated NK cells and GZMK⁺ resting NK cells, whereas no other NK cell subpopulations differed in colon cancer progression. KIR2DL4, also known as CD158d, is an unusual member of the killer cell Ig-like receptor family expressed in all NK cells (62 [↗](#)). Studies revealed that KIR2DL4 activates the cytotoxicity of NK cells (63 [↗](#)), and NK-92 cells stimulated with KIR2DL4 had higher cytotoxicity against breast cancer cells (64 [↗](#)). Consistent with previous studies, we also observed that the expression of the KIR2DL4⁺ GPR171⁺ activated NK cell subtype was decreased during CCLM progression, possibly reflecting the anti-tumor activity of this population. However, the GZMK⁺ resting NK cell subsets were first identified as potential tumor promoters. Moreover, it has been proven that GZMK^{high} CD8⁺ T effector memory cells, a cell subset that is particularly similar to GZMK⁺ NK cells, were associated with poor clinical outcomes in patients with colorectal tumors. Our results also showed that GZMK⁺ resting NK cells highly infiltrate the cancer region, and this predicts a worse prognosis in patients with colon cancer. However, despite the interesting and gratifying clinical relevance, the mechanism remains unknown.

Considering the evolutionary trajectory and the characteristic distribution of these metastasis-associated NK cell subtypes, we predicted that activated NK cells might differentiate into resting NK cells under the action of tumor cells. Consistent with a recent study that indicated that exposure to cancer cells causes NK cells to lose their cytotoxic ability (28), we identified the tendency of NK cells to differentiate to an resting state with decreasing expression of activated marker KIR2DL4 as well as increasing expression of decidual-like NK cells markers: CD9, CD49a and PD-1 (27), which are only triggered by cell-to-cell interactions as validated by FACS. It is worth mentioning that decidual-like NK cells were the newly identified subset of NK cells that promote tumors (27).

Additionally, NK-mediated tumor cell editing appears to partly depend on the release of cytokines in in vitro co-culture experiments. SCF, the natural ligand of c-Kit with membrane-bound (mSCF) and soluble forms (sSCF) (65), enhances the growth and migration of cancer cells such as cervical cancer (66), Ewing's sarcoma (67) and CRC (43). Although the dimeric mSCF is remarkably more active and induces a more persistent activation of the receptor (65, 68), the tumor-promoting effect of the supernatant from the co-culture system in the current study and the reversion effect of imatinib for the co-culture supernatant induced pro-oncogenic effects suggests an important role of sSCF on colon cancer cells. However, no up-regulation of KITLG (the SCF encoding gene) was found in HCT-116 after co-culturing with NK cells (Fig. S11A). In addition, when using DLD-1, a colon cancer cell with a nearly 15 times lower expression of KITLG than HCT-116 (Fig. S11B), we also did not observe co-cultured supernatant induced tumor-promoting effects. We thus hypothesized that in our experimental system, NK cells may promote the cleavage and release of the extracellular part of SCF on the tumor cell surface after direct contact with tumor cells, and thus increase the concentration of sSCF, which induces the malignant phenotype of colon cancer cells in a paracrine manner (Fig. 9). However, the detailed mechanism still needs to be further studied.

In summary, we have unveiled the spatial and cellular profiles of TME from tumors and paratumor tissues of CCLM. Our analysis uncovered the different states, functions, and dynamic nature of NK cells in different CCLM settings, which can be used for further identification of regulatory mechanisms and for developing potential therapeutic targets.

Data Availability

All data generated or analyzed during this study are available from the corresponding author on reasonable request.

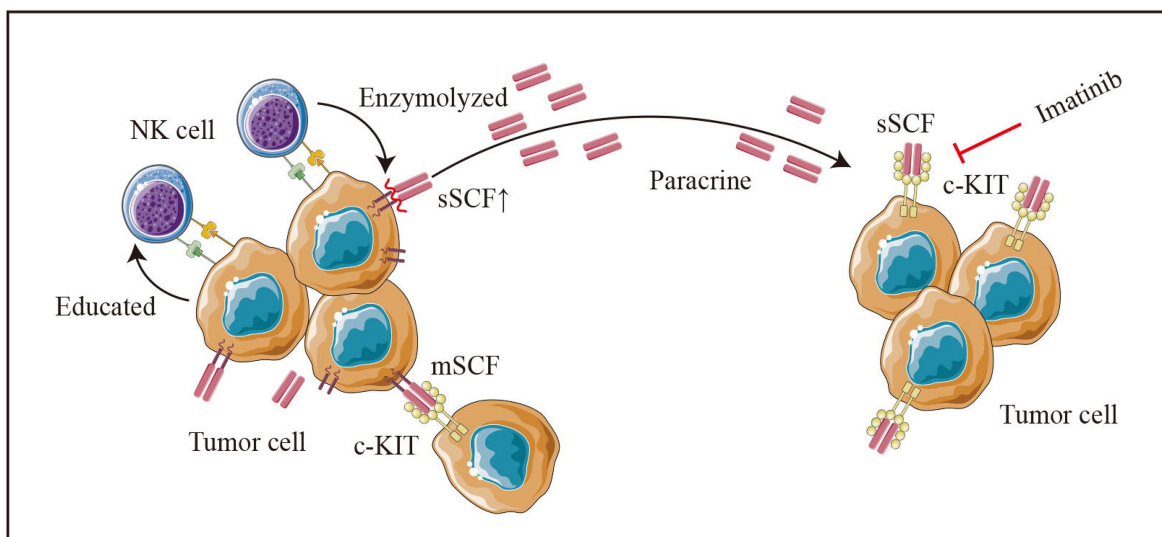


Figure.9.

Schematic diagram

Colon cancer cells (HCT-116 cells) educate NK cells as resting status depends on cell-to-cell interaction. Tumor-educated NK cells subsequent enhances tumor malignancy in a paracrine manner by elevating tumor-derived sSCF.

Funding

Funder	Grant reference number	Author
National Natural Science Foundation of China	32070151	Xian Shen
Key R&D Program of Zhejiang Province	2020C03029	Xiangyang Xue
Key R&D Program of Zhejiang Province	2021C03120	Xian Shen
the Zhejiang Provincial Natural Science Foundation Project	Q24H160147	Wangkai Xie
the Postdoctoral Fellowship Program of CPSF	GZC20231956	Wangkai Xie
the Wenzhou Municipal Science and Technology Bureau Program	Y2020215	Dan Xiang
The funders had no role in study design, data collection and interpretation, or the decision to submit the work for publication		

Acknowledgements

We thank Prof. Qiang Gao from Fudan University and Prof. Xiaoming Zhang from Shanghai Institute of Immunity and Infection, CAS for sharing the SC RNA sequencing and ST data. We thank Zhejiang Key Laboratory of New Techniques for Diagnosis and Treatment of Critical Diseases of Pancreas and Liver and Zhejiang International science and Technology Cooperation base for tumor transformation research for supporting this study. We thank Editage Group (<https://www.editage.cn/>) for polishing the draft of this manuscript.

1

Authors' Contributions

CCM and **YYC**: Conceptualization, experimental design and execution, data curation, method development, formal analysis, statistics, writing original draft, and editing the manuscript. **DX** and **TMZ**: conceptualization, data curation and method development. **YXL**, **CL** and **JYY**: experimental design. **MHL** and **TM**: experimental design. **WKX**, **DFM** and **ZH**: Validation. **DX**, **MDL** and **XS**: Experimental design, supervision, funding acquisition, and manuscript editing. **XYX** **XS**: Conceptualization, supervision, writing manuscript, project administration.

Supplemental Figures

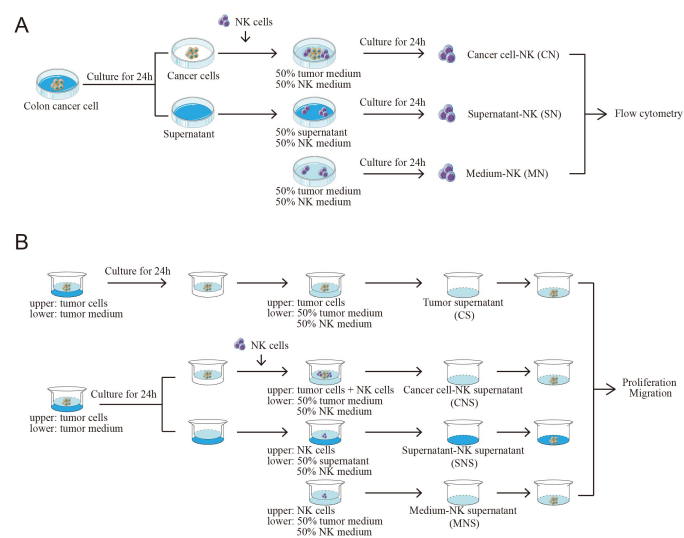


Figure S1.

Schematic overview of the in vitro experimental design.

A. Schematic overview of coculture experiments design. B. Schematic overview of coculture experiments design in transwells.

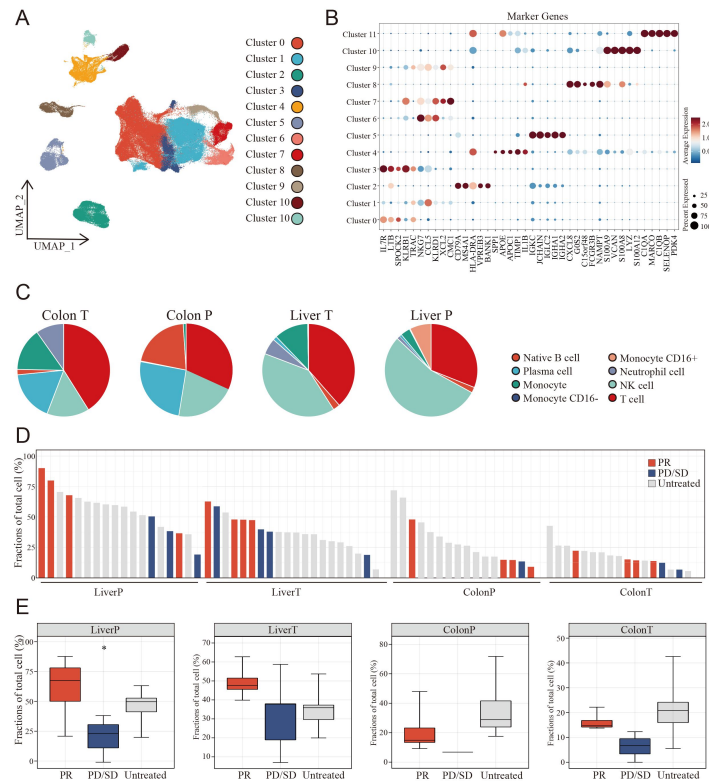


Figure S2.

Cluster characterization of the global landscape of CCLM revealed by single-cell transcriptomics.

A. The UMAP plot of all 12 cell clusters. B. Dot plots showing average expression of marker genes in indicated cell clusters. C. The pie chart of the proportion of main immune cells across tissues. D. The proportion of NK cells in total immune cells of each sample (including PR, PD/SD and untreated patients) from the Liver P, Liver T, Colon P, Colon T. E. Proportions of NK cells in PR, PD/SD and untreated patients among Liver P, Liver T, Colon P, Colon T groups.

Cellular identification in spatial transcriptomic samples.

A

UMAP_2
UMAP_1

Cluster 0
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5
Cluster 6
Cluster 7
Cluster 8
Cluster 9
Cluster 10

B

Marker Genes

Cluster 10
Cluster 9
Cluster 8
Cluster 7
Cluster 6
Cluster 5
Cluster 4
Cluster 3
Cluster 2
Cluster 1
Cluster 0

APOL1
BIP
ALB
COL1A1
COL4A1
COL6A1
PCSK
BMP4
LRR7A
INF3B
JCHAIN
CSF3B
VAMP1
SLEKOP
CKB
HSP90
AC3
CSRP1
MTORF1
MTORF2
TNNI3
IL1RA
SERPINE1
LITB
TRBC2
CORO1A

Average Expression
Percent Expressed

C

ALB
MYH11
COL1A1
CD24
KRT8
JCHAIN
CD79A
CD14
CXCL8
CD3D

Gene expression features of each sample. The feature plots showed the expression distributions of ALB, CD24, COL1A1, MYH11, CXCL8, CD14, CD79A AND CD3D in each spatial transcriptomic section.

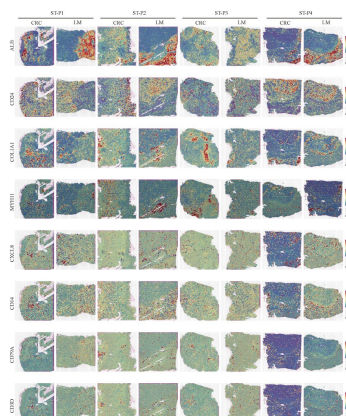


Figure S5.

Clinical relationship between NK cells subsets and colon cancer revealed by TCGA COAD cohort.

A. The relationship of activated and resting NK cell and clinical characteristics of colon cancer in TCGA COAD cohort. B. The relationship of 22 immune cells percentage determined by CIBERSORT and prognosis of colon cancer in TCGA COAD cohort.

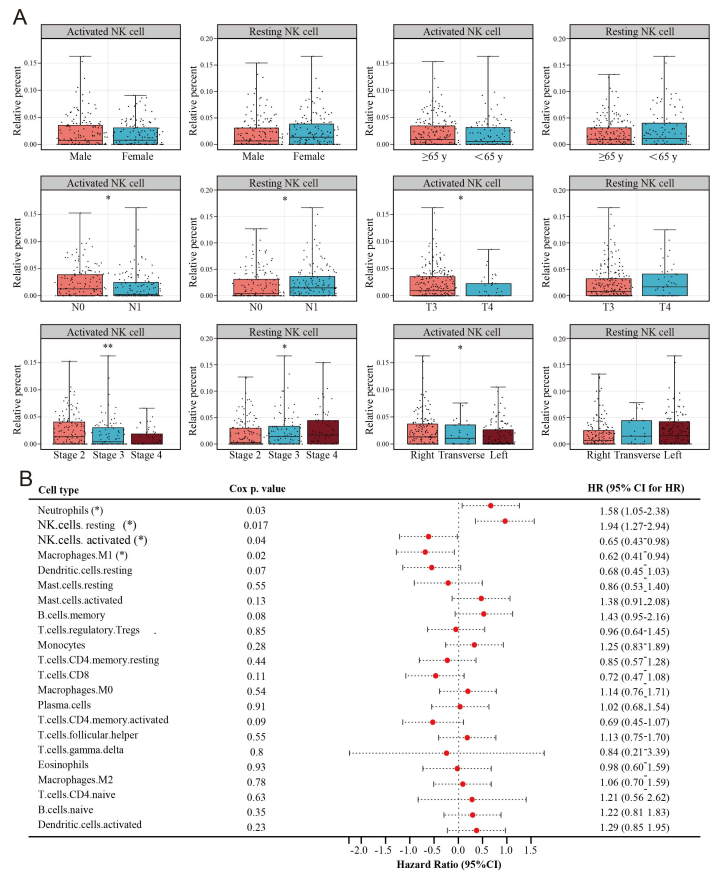


Figure S6.

The proportions of eight clusters of NK cells in Colon P, Colon T, Liver T and LN.

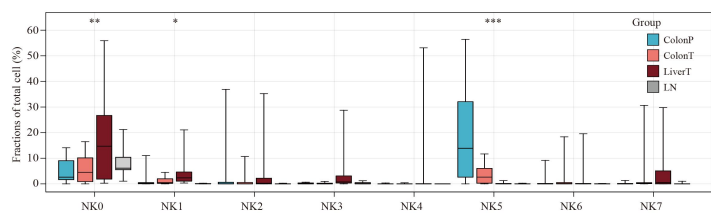


Figure S7.

Figure.S7 NK cell-mediated tumor promoting effect in colon cancer cells (DLD-1).

A. Clone formation assay showed the NK cell-mediated inductive effect on cell proliferation of DLD-1 cell between CNS and CS groups. B-C: CCK-8 assay showed the NK cell-mediated inductive effect on cell proliferation of DLD-1 cell between CNS and CS groups. D-F. The NK cell-mediated inductive effect on migration and invasion of DLD-1 cell between CNS and CS groups. G. CCK-8 assay showed the NK cell-mediated inductive effect on cell proliferation of DLD-1 cell among CNS, SNS, MNS groups. H-I. Clone formation assay showed the NK cell-mediated inductive effect on cell proliferation of DLD-1 cell among CNS, SNS, MNS groups. J-L The NK cell-mediated inductive effect on migration and invasion of DLD-1 cell among CNS, SNS, MNS groups.

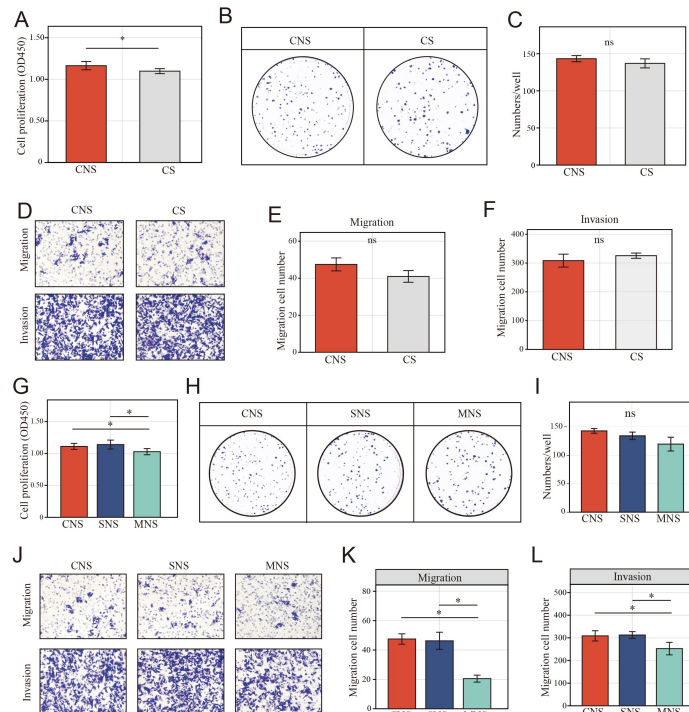


Figure S8.

The positive gate locations of CD56, GZMK, KIR2DL4, CD9, CD49a, PD-1 defined according to the FMO control.

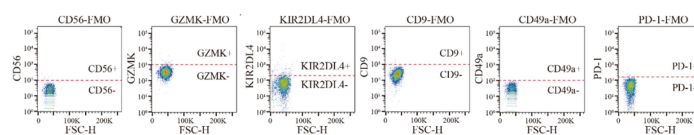


Figure S9.

Phenotype switch (CD56+, GZMK+) of NK cells was analyzed by FACS after fixation and permeabilization in different co-cultured groups.

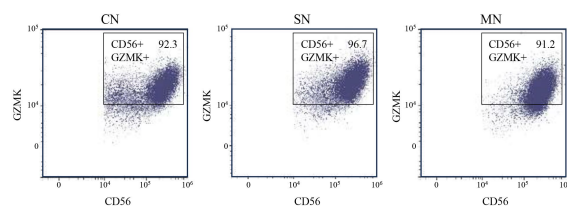
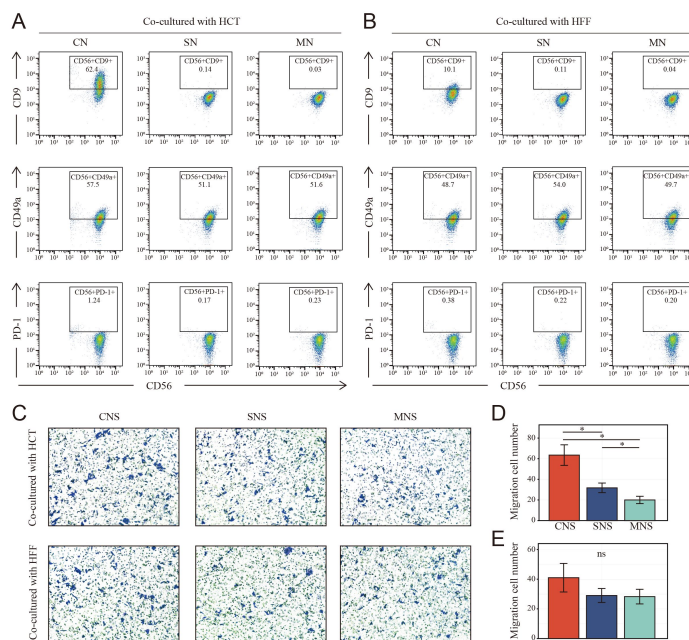


Figure S10.

Phenotype switch of NK cells in different co-cultured system and the corresponding NK cell-mediated effect on cell migration of fresh colon cancer cell (HCT-116).

A-B: NK cells underwent phenotype switch (high expression of CD9) when cocultured with HCT and HFF, the phenotype switch was more obvious when co-cultured with HCT. CN: NK cells cocultured with HCT/HFF; SN: NK cells cocultured with supernatant of HCT/HFF; MN: NK cells cocultured in fresh medium. C-E: Transwell assay showed the only tumor co-cultured NK mediated the inductive effect on cell migration of colon cancer cell (HCT-116). CNS: Colon cancer cells were cultured in the supernatant from co-culture system that NK and HCT/HFF were cultured in direct contact; SNS: Colon cancer cells were cultured in the supernatant from co-culture system that NK cocultured with supernatant of HCT/HFF; MNS: Colon cancer cells were cultured in the fresh medium.



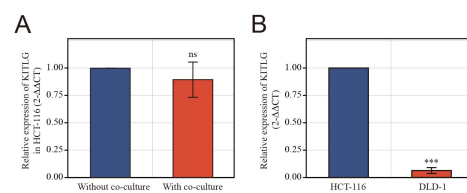


Figure S11.

Relative KITLG expression in different groups.

A. RT-qPCR analysis of KITLG expression in HCT-116 with and without co-cultured with NK cells. B. RT-qPCR analysis of KITLG expression in HCT-116 and DLD-1 cells.

References

1. Sung H, et al. (2021) **Global Cancer Statistics 2020: GLOBOCAN Estimates of Incidence and Mortality Worldwide for 36 Cancers in 185 Countries** *CA Cancer J Clin* **71**:209–249
2. Siegel RL, et al. (2020) **Colorectal cancer statistics, 2020** *CA Cancer J Clin* **70**:145–164
3. Wang F, et al. (2023) **Single-cell and spatial transcriptome analysis reveals the cellular heterogeneity of liver metastatic colorectal cancer** *Sci Adv* **9**
4. Chua TC, Saxena A, Chu F, Zhao J, Morris DL (2011) **Predictors of cure after hepatic resection of colorectal liver metastases: an analysis of actual 5-and 10-year survivors** *J Surg Oncol* **103**:796–800
5. Zampino MG, et al. (2016) **Treatments for colorectal liver metastases: A new focus on a familiar concept** *Crit Rev Oncol Hematol* **108**:154–163
6. Ruers T, et al. (2017) **Local Treatment of Unresectable Colorectal Liver Metastases: Results of a Randomized Phase II Trial** *J Natl Cancer Inst* **109**
7. Wancata LM, Banerjee M, Muenz DG, Haymart MR, Wong SL (2016) **Conditional survival in advanced colorectal cancer and surgery** *J Surg Res* **201**:196–201
8. Piawah S, Venook AP (2019) **Targeted therapy for colorectal cancer metastases: A review of current methods of molecularly targeted therapy and the use of tumor biomarkers in the treatment of metastatic colorectal cancer** *Cancer* **125**:4139–4147
9. Yao J, Chen Y, Lin Z (2023) **Exosomes: Mediators in microenvironment of colorectal cancer** *Int J Cancer* **153**:904–917
10. Liu H, Wang Z, Zhou Y, Yang Y (2023) **MDSCs in breast cancer: an important enabler of tumor progression and an emerging therapeutic target** *Front Immunol* **14**
11. Ngambenjawong C, Gustafson HH, Pun SH (2017) **Progress in tumor-associated macrophage (TAM)-targeted therapeutics** *Adv Drug Deliv Rev* **114**:206–221
12. Li S, et al. (2019) **Tumor-associated neutrophils induce EMT by IL-17a to promote migration and invasion in gastric cancer cells** *J Exp Clin Cancer Res* **38**
13. Tiberti S, et al. (2022) **GZMK(high) CD8(+) T effector memory cells are associated with CD15(high) neutrophil abundance in non-metastatic colorectal tumors and predict poor clinical outcome** *Nat Commun* **13**
14. Maskalenko NA, Zhigarev D, Campbell KS (2022) **Harnessing natural killer cells for cancer immunotherapy: dispatching the first responders** *Nat Rev Drug Discov* **21**:559–577
15. Zhong F, et al. (2022) **Innate tumor killers in colorectal cancer** *Cancer Lett* **527**:115–126
16. Villegas FR, et al. (2002) **Prognostic significance of tumor infiltrating natural killer cells subset CD57 in patients with squamous cell lung cancer** *Lung Cancer* **35**:23–28

17. Ishigami S, et al. (2000) **Prognostic value of intratumoral natural killer cells in gastric carcinoma** *Cancer* **88**:577–583
18. Lamb MG, Rangarajan HG, Tullius BP, Lee DA (2021) **Natural killer cell therapy for hematologic malignancies: successes, challenges, and the future** *Stem Cell Res Ther* **12**
19. Liu E, et al. (2020) **Use of CAR-Transduced Natural Killer Cells in CD19-Positive Lymphoid Tumors** *N Engl J Med* **382**:545–553
20. Zhang H, Yang L, Wang T, Li Z (2024) **NK cell-based tumor immunotherapy** *Bioact Mater* **31**:63–86
21. Hu W, Wang G, Huang D, Sui M, Xu Y (2019) **Cancer Immunotherapy Based on Natural Killer Cells: Current Progress and New Opportunities** *Front Immunol* **10**
22. Oh S, Lee JH, Kwack K, Choi SW (2019) **Natural Killer Cell Therapy: A New Treatment Paradigm for Solid Tumors** *Cancers* **11**
23. Bruno A, et al. (2013) **The proangiogenic phenotype of natural killer cells in patients with non-small cell lung cancer** *Neoplasia* **15**:133–142
24. Carrega P, et al. (2008) **Natural killer cells infiltrating human nonsmall-cell lung cancer are enriched in CD56 bright CD16(-) cells and display an impaired capability to kill tumor cells** *Cancer* **112**:863–875
25. Zhang QF, et al. (2017) **Liver-infiltrating CD11b(-)CD27(-) NK subsets account for NK-cell dysfunction in patients with hepatocellular carcinoma and are associated with tumor progression** *Cell Mol Immunol* **14**:819–829
26. Jin J, et al. (2013) **CD11b(-)CD27(-) NK cells are associated with the progression of lung carcinoma** *PLoS One* **8**
27. Albini A, Noonan DM (2021) **Decidual-Like NK Cell Polarization: From Cancer Killing to Cancer Nurturing** *Cancer Discov* **11**:28–33
28. Chan IS, et al. (2020) **Cancer cells educate natural killer cells to a metastasis-promoting cell state** *J Cell Biol* **219**
29. Huergo-Zapico L, et al. (2018) **NK-cell Editing Mediates Epithelial-to-Mesenchymal Transition via Phenotypic and Proteomic Changes in Melanoma Cell Lines** *Cancer Res* **78**:3913–3925
30. Wu Y, et al. (2022) **Spatiotemporal Immune Landscape of Colorectal Cancer Liver Metastasis at Single-Cell Level** *Cancer Discov* **12**:134–153
31. Hao Y, et al. (2021) **Integrated analysis of multimodal single-cell data** *Cell* **184**:3573–3587
32. Peng Z, et al. (2023) **Interactions between MFAP5 + fibroblasts and tumor-infiltrating myeloid cells shape the malignant microenvironment of colorectal cancer** *J Transl Med* **21**
33. Zheng H, Liu H, Ge Y, Wang X (2021) **Integrated single-cell and bulk RNA sequencing analysis identifies a cancer associated fibroblast-related signature for predicting prognosis and therapeutic responses in colorectal cancer** *Cancer Cell Int* **21**

34. Hong B, et al. (2022) **Single-cell transcriptional profiling reveals heterogeneity and developmental trajectories of Ewing sarcoma** *J Cancer Res Clin Oncol* **148**:3267–3280
35. Qiu X, et al. (2017) **Single-cell mRNA quantification and differential analysis with Censur** *Nat Methods* **14**:309–315
36. Goldman MJ, et al. (2020) **Visualizing and interpreting cancer genomics data via the Xena platform** *Nat Biotechnol* **38**:675–678
37. Chen B, Khodadoust MS, Liu CL, Newman AM, Alizadeh AA (2018) **Profiling Tumor Infiltrating Immune Cells with CIBERSORT** *Methods Mol Biol* **1711**:243–259
38. Aran D, Hu Z, Butte AJ (2017) **xCell: digitally portraying the tissue cellular heterogeneity landscape** *Genome Biol* **18**
39. Racle J, de Jonge K, Baumgaertner P, Speiser DE, Gfeller D (2017) **Simultaneous enumeration of cancer and immune cell types from bulk tumor gene expression data** *Elife* **6**
40. Becht E, et al. (2016) **Erratum to: Estimating the population abundance of tissue-infiltrating immune and stromal cell populations using gene expression** *Genome Biol* **17**
41. Yu G, Wang LG, Han Y, He QY (2012) **clusterProfiler: an R package for comparing biological themes among gene clusters** *OMICS* **16**:284–287
42. Liberzon A, et al. (2015) **The Molecular Signatures Database (MSigDB) hallmark gene set collection** *Cell Syst* **1**:417–425
43. Yasuda A, et al. (2007) **Stem cell factor/c-kit receptor signaling enhances the proliferation and invasion of colorectal cancer cells through the PI3K/Akt pathway** *Dig Dis Sci* **52**:2292–2300
44. Manfredi S, et al. (2006) **Epidemiology and management of liver metastases from colorectal cancer** *Ann Surg* **244**:254–259
45. Quail DF, Joyce JA (2013) **Microenvironmental regulation of tumor progression and metastasis** *Nat Med* **19**:1423–1437
46. Wei Q, et al. (2017) **Multiregion whole-exome sequencing of matched primary and metastatic tumors revealed genomic heterogeneity and suggested polyclonal seeding in colorectal cancer metastasis** *Ann Oncol* **28**:2135–2141
47. Yaeger R, et al. (2018) **Clinical Sequencing Defines the Genomic Landscape of Metastatic Colorectal Cancer** *Cancer Cell* **33**:125–136
48. Kumar S, et al. (2018) **DeltaNp63-driven recruitment of myeloid-derived suppressor cells promotes metastasis in triple-negative breast cancer** *J Clin Invest* **128**:5095–5109
49. Hanahan D, Coussens LM (2012) **Accessories to the crime: functions of cells recruited to the tumor microenvironment** *Cancer Cell* **21**:309–322
50. Zhou SN, et al. (2021) **Comparison of Immune Microenvironment Between Colon and Liver Metastatic Tissue in Colon Cancer Patients with Liver Metastasis** *Dig Dis Sci* **66**:474–482

51. Huntington ND, Voshchenrich CA, Di Santo JP (2007) **Developmental pathways that generate natural-killer-cell diversity in mice and humans** *Nat Rev Immunol* **7**:703–714
52. Freud AG, Caligiuri MA (2006) **Human natural killer cell development** *Immunol Rev* **214**:56–72
53. Cooper MA, Fehniger TA, Caligiuri MA (2001) **The biology of human natural killer-cell subsets** *Trends Immunol* **22**:633–640
54. Jacobs R, et al. (2001) **CD56bright cells differ in their KIR repertoire and cytotoxic features from CD56dim NK cells** *Eur J Immunol* **31**:3121–3127
55. Moretta L (2010) **Dissecting CD56dim human NK cells** *Blood* **116**:3689–3691
56. Tang F, et al. (2023) **A pan-cancer single-cell panorama of human natural killer cells** *Cell*
57. Xia T, et al. (2022) **Immune cell atlas of cholangiocarcinomas reveals distinct tumor microenvironments and associated prognoses** *J Hematol Oncol* **15**
58. Kim R, et al. (2022) **Early Tumor-Immune Microenvironmental Remodeling and Response to First-Line Fluoropyrimidine and Platinum Chemotherapy in Advanced Gastric Cancer** *Cancer Discov* **12**:984–1001
59. Laskowski TJ, Biederstadt A, Rezvani K (2022) **Natural killer cells in antitumour adoptive cell immunotherapy** *Nat Rev Cancer* **22**:557–575
60. Myers JA, Miller JS (2021) **Exploring the NK cell platform for cancer immunotherapy** *Nat Rev Clin Oncol* **18**:85–100
61. Thacker G, et al. (2023) **Immature natural killer cells promote progression of triple-negative breast cancer** *Sci Transl Med* **15**
62. Andre P, et al. (2001) **New nomenclature for MHC receptors** *Nat Immunol* **2**
63. Faure M, Long EO (2002) **KIR2DL4 (CD158d), an NK cell-activating receptor with inhibitory potential** *J Immunol* **168**:6208–6214
64. Kilic N, Dastouri M, Kandemir I, Yilmaz E (2023) **The effects of KIR2DL4 stimulated NK-92 cells on the apoptotic pathways of HER2 + /HER-breast cancer cells** *Med Oncol* **40**
65. Hsu YR, et al. (1997) **The majority of stem cell factor exists as monomer under physiological conditions. Implications for dimerization mediating biological activity** *J Biol Chem* **272**:6406–6415
66. Aguilar C, Aguilar C, Lopez-Marure R, Jimenez-Sanchez A, Rocha-Zavaleta L (2014) **Co-stimulation with stem cell factor and erythropoietin enhances migration of c-Kit expressing cervical cancer cells through the sustained activation of ERK1/2** *Mol Med Rep* **9**:1895–1902
67. Landuzzi L, et al. (2000) **The metastatic ability of Ewing's sarcoma cells is modulated by stem cell factor and by its receptor c-kit** *Am J Pathol* **157**:2123–2131
68. Langley KE, et al. (1993) **Soluble stem cell factor in human serum** *Blood* **81**:656–660

Editors

Reviewing Editor

Florent Ginhoux

Singapore Immunology Network, Singapore, Singapore

Senior Editor

Paivi Ojala

University of Helsinki, Helsinki, Finland

Reviewer #1 (Public Review):

Summary:

Mao and colleagues re-analysed published spatial, bulk and single-cell transcriptomic datasets from primary colorectal cancers and colorectal cancer derived liver metastases. The analyses of paired cancer and non-cancer tissue samples showed that T cells are enriched in tumour tissue, accompanied by a reduction in the fraction of NK cells in the cancer tissue transcriptional datasets. Furthermore, the authors show that tumour tissue has higher fraction of GZMK⁺ (resting) NK cells and suggested a correlation between the presence of these cells and poor prognosis for cancer patients. In contrast, the increased frequency of KIR2DL4⁺ (activated) NK cells correlates with improved survival of cancer patients.

Strengths:

Authors performed a comprehensive analysis of published datasets, integrating spatial and single-cell transcriptomic data, which allowed them to discover enrichment of GZMK⁺ NK cells in cancer tissues.

Weakness:

The authors provided insufficient experimental evidence to support their claim that GZMK⁺ NK cells contribute to worse prognosis for cancer patients or promote cancer progression. While one can visually observe an increased fraction of GZMK⁺ NK cells compared to KIR2DL4⁺ NK cells in cancer tissues, no quantification is shown. They did not present any preclinical (animal model) or clinical data suggesting a causal relationship between NK cells and tumour growth. Thus, while a correlation may exist between the presence of GZMK⁺ NK cells and poorer tumour prognosis, causation cannot be claimed based on the available evidence. Furthermore, the in vitro data provided is limited to a single NK cell line derived from a lymphoma patient, which does not fully represent the diversity and functionality of human NK cells.

<https://doi.org/10.7554/eLife.97201.2.sa2>

Reviewer #2 (Public Review):

Summary:

This manuscript investigates the role of the abundant NK cells that are observed in colon cancer liver metastasis using sequencing and spatial approaches in an effort to clarify the pro and anti-tumorigenic properties of NK cells. This descriptive study characterizes different categories of NK cells in tumor and tumor adjacent tissues and some correlations. An attempt has been made using pseudotime trajectory analysis but no models around how these NK cells might be regulated is provided.

Strengths:

This study integrates multiomics data to attempt to resolve correlates of protection that might be useful in understanding NK cell diversity and activation. The authors have strengthened the study in revision by demonstrating the very strong correlation between Granzyme+ NK cells and the poor prognosis, but the main claims are only partially supported.

Weaknesses:

While this work is interesting, the power of such studies are in taking the discovered information and applying this to other cohorts to determine the strength and predictive power of the genes identified. It is also clear that these 'snapshots' analysed poorly take account of the dynamic temporal changes that occur within a tumour. It would have been good to see a proposed model of NK cell regulation as it might occur in the tumour (accounting for turnover and recruitment) beyond the static data. Further evidence linking mechanistic causality to prognostic outcome would provide significant data for approaches forward.

<https://doi.org/10.7554/eLife.97201.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer#1:

Comment #1: It is unclear how the fraction of NK cell populations is quantified in the spatial-seq datasets. Figures display spatial data with expression scores, but the method for calculating the score and determining NK cell presence in tumor tissue is ambiguous. Clarification is needed on whether the identification relied solely on visual inspection or if quantitative analyses using other criteria were conducted.

Thank you for your questions. We removed the background and made the accordingly modifications according to your demand. We used the AddModuleScore function in Seurat to quantify the main immune subpopulations in spatial-seq using the gene sets identified in single-cell-seq. Additionally, the tumor and non-tumor region was identified by immunohistochemistry as well as cell clusters in spatial-seq, it is rough that we can't quantify the NK cell presence in each region precisely. The consolation is that the differences of NK cell presence in tumor and non-tumor region is observable by visual inspection. The methodology has been supplemented in the revised manuscript (line 190-193).

Comment #2: The authors do not provide a clear definition of "resting" NK cells. It remains unclear whether they refer to a senescent state or a non-matured NK cell population. Furthermore, the criteria used to define resting and activated cells based on the expression of KIR2DL4, GPR183, GRP171, CD69, IFNG, GZMK, TTC38, CD160, and PLEKHF1 in Figure 4 are not well-defined. The expression patterns of these genes in Figure 4D are not distinct, and it is unclear which combination of genes was used to classify the populations. Clarification is needed on whether the presence of GZMK alone defines resting NK cells, or if the presence of any of the described genes (GZMK, TTC38, or CD160) is sufficient. Additionally, the method used for this classification, whether visual or algorithm-based, should be described.

Thank you for your question. The resting and activated NK cells was defined by the preferential expression of the described resting genes (AZU, BPI, CAMP, CD160, CD2, CDHR1,

CEACAM8, DEFA4, ELANE, GFI1, GZMK, KLRC4, MGAM, MS4A3, NME8, PLEKHF1, TEP1, TRBC1, TTC38, ZNF135) and activated NK genes (APOBEC3G, APOL6, CCL4, CCND2, CD69, CDK6, CSF2, DPP4, FASLG, GPR171, GPR18, GRAP2, IFNG, KIR2DL4, KIR2DS4, LTA, LTB, NCR3, OSM, PTGER2, SOCS1, TNFSF14) in CIBERSORT. Actually, these marker genes were not specifically expressed in a single NK cells subset. On the other hand, combined with further flow cytometric analysis verification, the resting NK cell tend to be a decidual-like NK cells and tumor- infiltrated NK cells with higher expression of CD9, CD49a and PD-1.

Comment #3: Criteria used to define high or low NK cell presence/infiltration in Figure 5 are not described in the main text or figure legend. Since, the claim that the presence of the resting or activated NK cells predicts cancer prognosis is based on this figure, this needs to be clearly described.

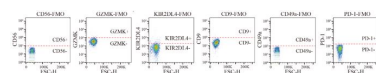
Thank you for your questions. The activated and resting NK cell percentage in TCGA and GSE29623 was determined by CIBERSORT. Additionally, the infiltration of activated and resting NK cell was also determined by the AddModuleScore function using the gene sets of activated and resting NK cell identified in single-cell-seq, the differences of activated and resting NK cell presence in tumor and non-tumor region is also determined by visual inspection. We have amended in the main text and figure legend in the revised manuscript.

Comment #4: The absence of FMO controls for KIR2DL4 or GZMK and the lack of increase in GZMK expression during co-culture with tumour lines raises concerns since GZMK was used as a defining feature of resting NK cells.

Thank you for your questions. We did a new batch of flow experiments and FMO controls of all the markers used in the experiments were set up to define the precise positive gate locations.

Author response image 1.

The positive gate locations of CD56, GZMK, KIR2DL4, CD9, CD49a, PD-1 defined according to the FMO control.



Comment #5: All the co-cultures were performed with tumour cell line only and no healthy cells, such as human foreskin fibroblasts, were used as control. In the absence of a non-tumour cell line, it is very difficult to draw any conclusions. Furthermore, to claim that resting or activated NK cells are responsible for tumour migration or proliferation, it is important to at least isolate resting and activated NK cells ex vivo and culture with tumour lines, instead of NK cell lines.

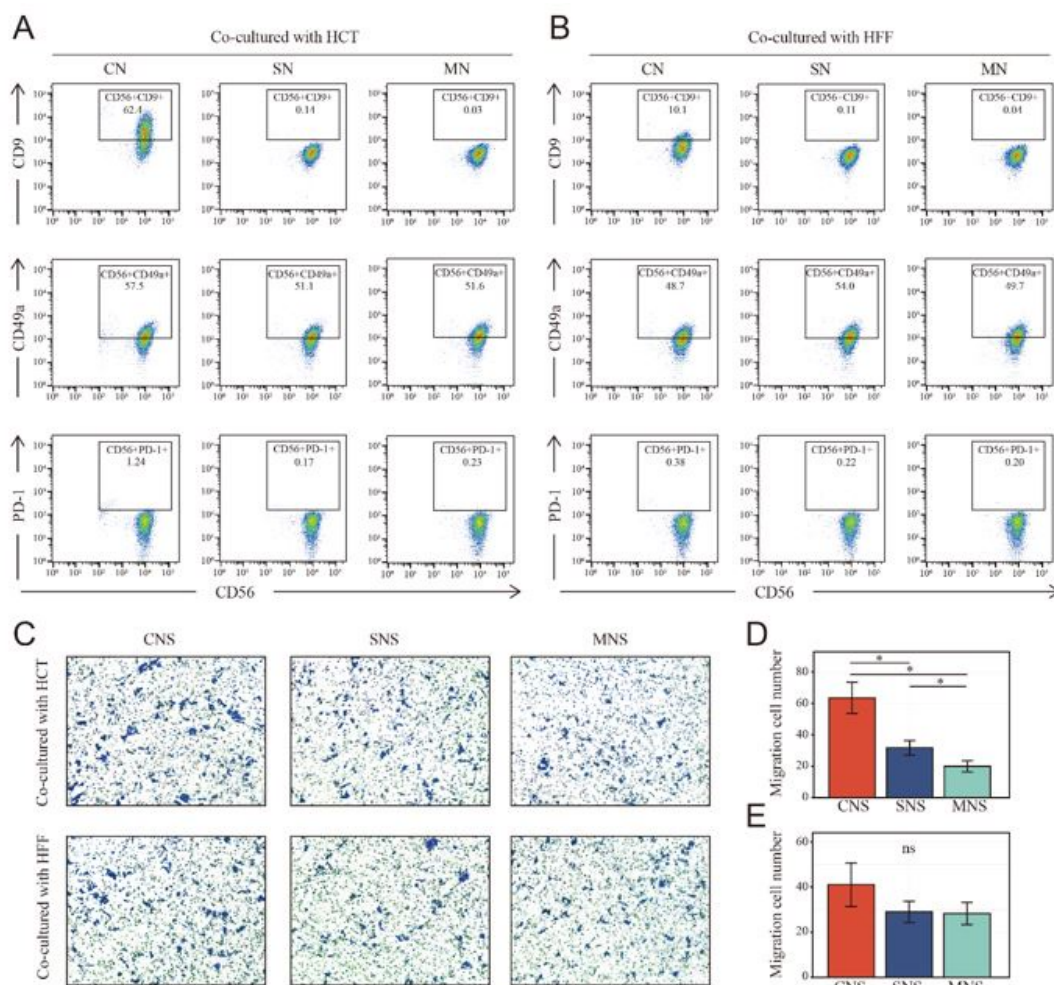
Thank you for your questions. According to your suggestion, NK cells were co-cultured with human foreskin fibroblasts, the phenotype was identified by Flow cytometry. When co-cultured with HFF in direct contact (CN group), NK cells were also tending towards tissue infiltration state (high expression of CD9). However, the domestication effect is significantly reduced compared to co-culturing with tumor cells. Additionally, unlike supernatant of CNS group (NK and HCT were in contact) from NK and HCT co-culture system could significantly increase the migration of fresh HCT, fresh HCT underwent a limited increase (no statistical significance was found) in migration when cultured in the supernatant from the co-culture system in which NK and HFF were in contact (CNS group), but not when co-cultures were performed in the cell supernatant (SNS group) and fresh medium (MNS group). Finally, we

tried to isolate resting and activated NK cells from fresh colon cancer surgical specimen. Unfortunately, the NK cells were too few to perform further functional experiments such as migration and proliferation.

Author response image 2.

Phenotype switch of NK cells in different co-cultured system and the corresponding NK cell-mediated effect on cell migration of fresh colon cancer cell (HCT-116).

A-B: NK cells underwent phenotype switch (high expression of CD9) when cocultured with HCT and HFF, the phenotype switch was more obvious when co-cultured with HCT. CN: NK cells cocultured with HCT/HFF; SN: NK cells cocultured with supernatant of HCT/HFF; MN: NK cells cocultured in fresh medium. C-E: Transwell assay showed the only tumor co-cultured NK mediated the inductive effect on cell migration of colon cancer cell (HCT-116). CNS: Colon cancer cells were cultured in the supernatant from co-culture system that NK and HCT/HFF were cultured in direct contact; SNS: Colon cancer cells were cultured in the supernatant from co-culture system that NK cocultured with supernatant of HCT/HFF; MNS: Colon cancer cells were cultured in the fresh medium.

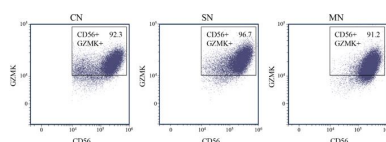


Comment #6: It seems that flow cytometric analyses and GZMK and KIR2DL4 staining were performed without cell permeabilization. Could authors confirm if this is accurate, or if they performed intracellular staining instead?

Thank you for your questions. For GZMK, which known as the secretory protein, flow cytometric analyses were performed both with (Fig.3) and without cell fixation and permeabilization, no significant differences were found among each group. The difference is that GZMK was nearly all negative without fixation and permeabilization while it is all positive with fixation and permeabilization. Conditions of flow cytometry analyses for GZMK may need further optimization or GZMK may not be a suitable flow cytometric marker for resting NK cells. On the other hand, for membrane protein such as CD56, CD9, CD49a, KIR2DL4, PD-1, staining was performed without cell permeabilization.

Author response image 3.

Phenotype switch (CD56+, GZMK+) of NK cells was analyzed by FACS after fixation and permeabilization in different co-cultured groups. CN: NK cells cocultured with colon cancer cells; SN: NK cells cocultured with supernatant of cancer cells; MN: NK cells cocultured in fresh medium.



Comment #7: The identity of the published datasets used for analysis is not provided, and references are not cited in the results section.

Thank you for your questions. We are sorry for the neglect of our previous work. We have added the information in the revised manuscript (section of Materials and Methods) (Line 123-128).

Comment #8: References are difficult to locate, as the main text follows APA style while the reference section is organized numerically with no clear order.

Thank you for your questions. We have modified the format of the references in the revised manuscript.

Comment #9: Figure 3 shows volcano plots showing DEG genes between tumor and healthy tissue NK cells are not described clearly, and authors did not discuss the significance of these genes, highlighted in the plot.

Thank you for your questions. Volcano plots of Figure 3 showed the DEGs between colon cancer with metastasis and without metastasis in TCGA database. We focused on the genes which were enriched in the pathway of “Natural killer cell mediated cytotoxicity” and found nearly all the genes enriched in the pathway were down-regulated in the colon cancer with metastasis. We have modified the description in the result section and added the description of importance of these genes in the discussion section in the revise manuscript (Line 322-326).

Comment #10: The meaning of "M0" and "M1" in Figures 5A and 5B is unclear and should be defined in the text.

Thank you for your questions. "M0" and "M1" in Figure 5A and 5B means “colon cancer without metastasis” and “colon cancer with metastasis”, respectively. We have modified in the revise manuscript (Line 350-354).

Comment #11: Terms such as "dynamic remodelling of NK cells" and "landscape of NK cells" are used without explanation, necessitating clarification of their meaning.

Thank you for your questions. We have modified in the revise manuscript (Line 331-334).

Comment #12: In vitro assays are described vaguely, making it difficult for readers to understand. More clarity is needed in describing these assays.

Thank you for your questions. We have added clarification in the revise manuscript (Line 205-211).

Reviewer #2:

Comment #1: This manuscript investigates the role of the abundant NK cells that are observed in colon cancer liver metastasis using sequencing and spatial approaches in an effort to clarify the pro and anti-tumorigenic properties of NK cells. This descriptive study characterises different categories of NK cells in tumor and tumor-adjacent tissues and some correlations. An attempt has been made using pseudotime trajectory analysis but no models around how these NK cells might be regulated are provided.

Thank you for your questions. The single-cell sequencing data enrolled in this study are CD45 positive immune cells and do not involve tumor cells, cellular communication analysis between NK cells and tumor cells cannot be conducted. The change process of NK can only be predicted through pseudotime trajectory analysis. Our hypothesis is that tumor cells domesticate NK cells into a tumor- infiltrated NK cells through direct contact, and flow cytometry experiments have also confirmed that tumor cells can only have such domestication through direct contact with NK cells (with prominent high expression of CD9). However, the detailed mechanism remained unclear.

Comment #2: A small number of patients are analyzed in this study. The descriptive gene markers, while interesting, need to be further validated to understand how strong this analysis might be and its potential application.

Thank you for your questions. The sample size included in this study is indeed a bit small, which is also a limitation of our study. However, this is the only large sample single-cell sequencing dataset could be found that includes primary colon cancer tissues, paired paratumor normal colon tissues, paired liver metastatic cancer tissue, and paired paratumor normal liver tissues. We will expand the sample size to further verify the current conclusion in subsequent experiments. In addition, the marker genes of different NK groups used in this study refer to the CIBERSORT's classification of activated NK cells and resting NK cells, which is a widely recognized indicator. We will verify the expression and clinical application value of the screened genes in tissues in subsequent studies.

Comment #3: Figure 1C and other figures throughout the paper. It is not clear how marker genes were selected.

Thank you for your questions. The marker genes displayed in the Figure.3C were the highly variable genes of each cell group as well as the marker genes of each immune cells, such as T cells (CD3D, CD3E), NK cells (NKG7, KLRD1), monocytes (LYZ, S100A8, S100A9), B cells (CD79A), plasma cells (JCHAIN, IGHA1, IGHA2), Neutrophils (CXCL8, FCGR3B).

Comment #4: Figure 1E. P and T have not been defined. Lines should not connect the datasets as they are independent assessments.

Thank you for your questions. P and T means paratumor normal tissues and tumor tissues, respectively. Which have been added in the caption of Figure 1E. Additionally, the single cell sequencing samples included in the study were paired, with primary colon cancer tissues, paired normal tissues adjacent to colon cancer, paired liver metastatic cancer tissue, and paired normal liver tissues from 20 colon cancer patients with liver metastasis, paired test analysis was thus performed.

Comment #5: Figure 2C. It is unclear what ST-P1 means. This is not a particularly informative figure.

Thank you for your questions. We are sorry that it was our annotation error. Actually, it is the spatial transcriptome of the primary colon cancer tissue and liver metastasis tissue of four patients. We have made the modifications in the revised manuscript.

Comment #6: Multiple figures - abbreviations are used but not provided in the legend. They occur in the text but are not directly related to the figures where they are used to label axes or groups.

Thank you for your questions. We have rechecked and made corresponding modifications in the revised manuscript.

Comment #6: Patients: it is not clear what other drugs patients have been exposed to or basic data (sex, age, underlying conditions etc)

Thank you for your questions. The baseline data of the patient of SC dataset and ST dataset were showed in the Table.1 and Table.2 followed, respectively. They were not presented before as no patients characteristics related analysis was performed in the current study.

Author response table 1.

The baseline data of patient from single cell sequencing database.

Patient	Age	Sex	TNM	Neoadjuvant chemotherapy
P1	71	Male	T1N0M1	N
P2	78	Female	T3N1M1	N
P3	60	Female	T2N0M1	N
P4	26	Female	T3N1M1	N
P5	64	Male	T2N0M1	Y (PR, FOLFOX)
P6	71	Male	T3N1M1	N
P7	52	Female	T3N1M1	Y (PR, XELOX)
P8	53	Male	T3N0M1	N
P9	35	Male	T1N0M1	Y (PR, XELOX)
P10	66	Male	T3N1M1	N
P11	43	Female	T3N1M1	N
P12	68	Male	T3N1M1	N
P13	69	Female	T3N1M1	Y (PD, XELOX)
P14	46	Female	T3N1M1	Y (PD, XELOX)
P15	51	Female	T3N1M1	Y (PR, FOLFOX)
P16	64	Female	T3N1M1	N
P17	49	Female	T3N1M1	N
P18	55	Male	T3N1M1	Y (PD, XELOX)
P19	63	Female	T2N0M1	Y (PD, XELOX)
P20	54	Male	T3N0M1	Y (SD, FOLFOX)

Author response table 2.

The baseline data of patient from spatial transcriptome database.

Patient	Age	Sex	TNM	Treatment
ST-P1	73	Male	T3N0M1	N
ST-P2	67	Female	T3N1M1	N
ST-P3	72	Male	T3N1M1	Y (XELOX)
ST-P4	68	Female	T3N1M1	Y (XELOX)

<https://doi.org/10.7554/eLife.97201.2.sa0>